
Hydra MAGE: Learning an Effective Theory of Superconductivity from Composition Alone

James Conde
jamesconde07@gmail.com
July 14, 2026

Contents

1	Introduction	2
2	Dataset	5
2.1	Training Data	5
2.2	Data Pipeline	6
2.3	Why Include Non-Superconductors?	6
2.4	Holdout Set	6
3	Methods	6
3.1	Architecture Overview	6
3.2	The 2048-Parameter Effective Theory	9
3.3	Training	9
3.3.1	Closing the Teacher-Forcing Gap with SCST	10
3.3.2	Physics Z Losses	11
3.3.3	Curriculum and Scheduling	11
3.3.4	Training Progression	11
3.3.5	Phase-Transition Heads (Curie MVP)	12
4	Results	13
4.1	T_c Prediction Performance	13
4.2	Latent Space Analysis	14
4.3	Per-Family Gradient Analysis	15
4.4	Extracting the Effective Theory as Equations	16
4.5	Anatomy of the Latent Space	17
4.6	An Emergent Phase-Transition Energy Scale	19
4.7	Generative Holdout Recovery	20
4.7.1	Error Analysis	21

4.8	External validation: joint composition-and- T_c recovery on 429 never-seen superconductors	21
4.9	Steering the Latent Space	23
4.10	Causal Validation: From Correlation to Control	24
4.11	What the T_c readout computes: a stored coordinate, not a mechanism	25
4.12	Sparse Decomposition of the Latent Space	27
4.13	Confident Failures Off the Manifold	27
4.14	From Latent Discovery to Deployable Candidates	29
5	Discussion and Future Work	29
6	Conclusion	33

Abstract

A neural network trained on scientific data can come to represent more physics than it was asked to learn. This paper presents Hydra MAGE, a multi-task autoencoder whose single 2048-dimensional latent space is read out by a dozen task-specific heads—the many-headed (“hydra”) design that names it—together with a mechanistic-interpretability study of what that latent encodes. The model jointly reconstructs composition, superconducting T_c , and material family from the shared latent; its core is trained on a balanced 60,570-material superconductor/non-superconductor set, with larger label-specific corpora and vocabulary-expansion negatives added for the auxiliary heads and the decoder. Its most striking behavior is generative: across a held-out set of 429 superconductors drawn from databases disjoint from training and scored jointly on both composition and T_c , latent-space search recovers all 429 with correct elements and near-correct stoichiometry—close enough for an experimentalist to identify and synthesize the material, with zero hallucinated elements—the final few targets closed by a validity-head successor of near-identical latent geometry together with differentiable T_c -gradient steering. Recovery holds across the full extent of the manifold, including its most sparsely-sampled regions and not only a densely-sampled core. The model also predicts phase transitions it was never trained on, forecasting ferromagnetic Curie temperatures of unseen magnets (several exactly); this motivated extending the architecture with heads that read further chemical properties from the same latent. Opening the model up, I extract per-family T_c regularities as auditable closed-form fits, map the anatomy and causal structure of the latent, and port the language-model interpretability toolkit—sparse autoencoders, activation steering, probes with control tasks, and feature-circuit ablation—to this scientific setting, using those artifacts to steer generation, which yields a cross-family shortlist of database-novel candidates that a per-candidate literature search resolves largely into rediscoveries of known superconducting chemistry (companion supplement)—leaving a single deeply-vetted survivor, Hf-doped FeSe ($T_c \approx 21$ K predicted), the one carried through a full literature search and found absent from $\sim 365,000$ database entries and the published record. Causal intervention also delimits the account: the T_c readout behaves as a stored coordinate of a joint (composition, T_c) representation rather than a physical mechanism. I characterize the result as a 2048-parameter effective theory of superconductivity spanning 14 material families—extracted, audited, and delimited.

1 Introduction

Discovering new superconducting materials remains a central challenge in condensed matter physics. Since Onnes’ discovery of superconductivity in mercury in 1911 (80), the field has progressed through elemental superconductors, intermetallic compounds (Nb_3Sn , Nb_3Ge), the cuprate revolution

beginning with Bednorz and Müller’s 1986 discovery of La-Ba-Cu-O (1), iron pnictides (2), MgB_2 (3), and most recently high-pressure hydrides where H_3S achieves $T_c = 203$ K at 155 GPa (4) and LaH_{10} reaches 250–260 K at 170–200 GPa (5). Each major family was discovered largely through serendipity or chemical intuition rather than systematic prediction—a pattern noted across the history of the field (6).

Superconductivity presents two intertwined holy grails. The *applied* goal is high-temperature superconductivity at ambient pressure, which would transform energy transmission, computing, and transportation. The *theoretical* goal is finding the equations that unify the diverse physical mechanisms giving rise to superconductivity. A unifying description *must* exist: all superconductors, regardless of mechanism, exhibit the same empirical phenomena—zero electrical resistance, the Meissner effect, and flux quantization in units of $h/2e$ (8). These universal properties demand a universal theoretical framework. The profound challenge is that the microscopic routes to these shared phenomena are remarkably diverse—phonon-mediated BCS pairing (7), *d*-wave spin-fluctuation pairing in cuprates (8), $s_{\pm}$ -wave inter-band pairing in iron pnictides (2), two-gap strong coupling in MgB_2 (3), and heavy-fermion/unconventional pairing—spanning vastly different energy scales, symmetries, and chemistry. These two goals are deeply connected: without a complete theory, it is unknown whether ambient-pressure room-temperature superconductivity is physically achievable or whether fundamental constraints—analogueous to the $T_c/J \approx 0.04$ – 0.07 bound observed across unconventional superconductors (9)—impose an upper limit that current materials are already approaching. A complete theory unifying these mechanisms would have implications far beyond materials science. Superconductivity has already had one such transformative impact: Anderson’s demonstration that the photon acquires an effective mass inside a superconductor through spontaneous symmetry breaking (10) served as the direct prototype for the Higgs mechanism, which explains how gauge bosons acquire mass in the Standard Model of particle physics. A deeper understanding of what connects the diverse manifestations of superconductivity—the universal principle that the empirical phenomena demand—could yield similarly far-reaching theoretical insights.

A single model trained across all known superconductor families is, implicitly, an attempt to learn this unifying physics from data. If the model organizes its latent space into a manifold where diverse mechanisms coexist with smooth transitions and shared geometric structure, it provides evidence—not proof, but evidence—that the superficially different microscopic mechanisms are connected by deeper commonalities that a sufficiently expressive representation can capture. The combinatorial space of possible compositions exceeds 10^{10} candidates (42), motivating computational approaches that can learn the implicit physics governing superconductivity and use it to navigate this space efficiently.

Materials informatics has made significant progress in predicting critical temperature (T_c) from chemical composition using machine learning. Hamidieh (11) demonstrated XGBoost models on SuperCon data; Stanev et al. (12) achieved $R^2 \approx 0.88$ with random forests and Magpie features (13); more recent work by Jung et al. (14) reached $R^2 = 0.952$ using gradient-boosted feature selection. However, these composition-only methods face a fundamental ceiling: a chemical formula is an incomplete specification of a superconductor. Many variables that critically determine T_c —crystal structure, oxygen ordering, apical oxygen distances, pressure, defect chemistry—are absent from composition-only databases (15). In cuprates, T_c can vary by up to a factor of 10 between materials with the same in-plane electronic structure but different out-of-plane geometry (16; 17). The SuperCon database, the largest repository of superconductors, records only chemical formulas; structural data is sparsely recorded and full 3D crystal structures are never given (15). Adding structural information via SOAP descriptors improves accuracy ($R^2 = 0.929$) but requires crystal-structure data unavailable for most superconductors (22).

Generative approaches have begun to address the inverse design problem—proposing new candidate materials rather than merely predicting properties of known ones. Wines et al. (18) combined a crystal diffusion variational autoencoder (CDVAE) with ALIGNN to generate and screen superconductor candidates. Zhong et al. (19) used a VAE-GAN hybrid for high- T_c composition generation. ScGAN (20) achieved a 23-fold improvement in superconductor discovery rate over random search. However, these approaches are typically two-stage pipelines: one model generates candidates, a separate model screens them for properties. Related generative efforts include diffusion models (SuperDiff (23)) and GANs for crystal-structure prediction (24). Across all of them the generative and predictive capabilities remain architecturally disconnected.

Sequence-based chemical generation. The Chemical VAE (25) demonstrated that joint training of a VAE with a property predictor produces latent spaces organized by property values. Materials-specific generative models have since been developed for solid-state inverse design (26) and composition space sampling (27). The pointer-generator mechanism (28) enables copying tokens from input sequences, which I adapt for exact element symbol reproduction. The fraction-aware tokenization extends these approaches to handle the decimal and fractional stoichiometries common in inorganic superconductors.

Mechanistic interpretability. A now-substantial literature reverse-engineers neural networks’ internal computations, developed almost entirely on language models: sparse autoencoders that decompose activations into monosemantic features (29; 30), feature-level causal circuits that can be traced and edited (31), activation steering and top-down representation reading and control (32; 33), and probing for emergent internal world models—the paradigm case being a sequence model that latently represents a game board it was never given (34). The field has matured toward its own controls and cautions: control-task baselines for probes (35), and stress tests showing that a component’s necessity or linear-decodability does not by itself establish a mechanistic role (36), a caution rooted in the superposition geometry underlying such distributed encodings (37). These tools have rarely left the language domain; a recent exception applies sparse autoencoders to speech recognition (38). This paper ports the toolkit—SAEs, activation steering, linear probes with control tasks, and causal ablation—to a scientific *materials* model, recovers the same phenomena (superposition, distributed directions, off-manifold overconfidence), closes the loop to generation, and—for one diagnosed failure mode—traces and edits an explicit feature circuit (Section 4.13), albeit at the readout level rather than as a full input-grounded causal graph.

Mechanistic interpretability of scientific models. Symbolic regression—in particular PySR (39)—has been used to distill trained networks into closed-form expressions, turning learned weights into human-auditable equations. A cautionary result by Vafa et al. (40) shows that transformers trained on planetary dynamics can achieve excellent predictive accuracy while failing to recover Newton’s law of gravitation: high fit quality does not by itself establish that a network has internalized the causal physics. Section 4.4 applies symbolic regression to Hydra MAGE’s T_c head and treats the Vafa et al. warning as a first-class caveat, including a documented retraction of one initially-celebrated extracted form. Element-substitution priors from large-scale materials discovery efforts such as GNoME (41) inform one of the generation strategies in Section 4.9.

Phase-transition temperature prediction. Beyond superconductivity, machine learning has been applied to ferromagnetic Curie temperature prediction; the strongest published composition-based result is a gradient-boosted model with 38.8 K test MAE (43). Section 4.6 documents Hydra MAGE spontaneously predicting Curie temperatures without magnetic supervision, and a trained Curie head that approaches this state-of-the-art (SOTA) while sharing a single latent space with all superconductivity tasks.

In this work, I present Hydra MAGE, a unified multi-task encoder-decoder (21) where the same 2048-dimensional latent space simultaneously supports formula generation and property prediction through shared decoder pathways. The architecture introduces several innovations: (1) a three-branch encoder combining learned element representations, Magpie compositional features, and T_c conditioning; (2) two complementary formula decoders—an autoregressive pointer-generator with fraction-aware tokenization, and a two-stage decoder that predicts the element set and its stoichiometric fractions in separate passes—whose disagreements flag unreliable generations; (3) jointly optimized decoder heads that create self-consistency learning channels shaping the latent space; and (4) a feedback mechanism where predicted T_c conditions the formula decoder through cross-attention memory tokens. The decoders are trained on 60,570 materials with a deliberately balanced SC/non-SC split; the T_c head—active only where the classifier admits a superconductor—reaches $R^2 = 0.9999$, and systematic latent-space search recovers 429 held-out superconductors on both composition and T_c .

Beyond predictive and generative performance, the central contribution of this paper is a *model-internals* program: extracting the effective theory the network learned as explicit per-family equations, mapping the anatomy of its latent space with gradient-based probes, documenting an emergent capability the model was never supervised on (a general phase-transition energy scale), and closing the loop by using those interpretability artifacts to steer generation toward novel candidate materials.

2 Dataset

2.1 Training Data

The model is trained on a composite dataset substantially larger and more diverse than the SuperCon subset used in prior work. Table 1 gives its full composition: an initial ingestion of 56,313 materials from six databases, followed by targeted vocabulary-expansion and boundary-negative augmentation added during training, bringing the corpus to 60,615 examples. The model trains on 60,570 of these; the remainder are holdout superconductors that appear in the source data and are withheld before training.

Table 1: Full training dataset composition. Six source databases contribute the initial 56,313-material ingestion; targeted vocabulary-expansion and boundary-negative augmentation added during training brings the corpus to 60,615 examples (26,178 SC / 34,437 non-SC). The model trains on 60,570 of these, withholding the holdout superconductors present in the source data.

Component	Source	Count
<i>Superconductors (initial ingestion)</i>		
SuperCon SC	NIMS (44)	16,521
NEMAD SC	NEMAD (45)	6,930
JARVIS hydrides SC	JARVIS-DFT (46)	2,740
<i>Subtotal SC</i>		26,191
<i>Non-superconductors (initial ingestion)</i>		
Materials Project non-SC	MP (47)	14,040
NEMAD non-SC	NEMAD magnetic/thermo/aniso	9,154
MP targeted non-SC	Rare-element neighborhoods	3,500
Family-labeled & other non-SC	SuperCon / mixed sources	3,428
<i>Subtotal non-SC</i>		30,122
<i>Vocabulary-expansion & boundary negatives (added in training)</i>		
Boundary negatives	MP, SC-adjacent compositions	3,416
Element-coverage (MP)	MP rare-element compounds	610
Element-coverage (synthetic)	Kr, Rn, Fr, Ra, Po, At, actinides	284
Doping-boundary entries	SC/non-SC transition series	9
Deduplication / cleanup		−17
<i>Subtotal augmentation</i>		+4,302
Grand Total (final training CSV)		60,615

All materials pass through a uniform pipeline: pymatgen (48) parsing, 145 Magpie features via matminer (49), and 6 L_p -norm features, producing consistent feature vectors per sample.

Dataset growth to 60,615 examples. Three targeted augmentations were added during extended training: (1) *boundary negatives*—3,416 non-superconducting compositions adjacent to known superconductors in composition space, added to sharpen the SC/non-SC decision boundary; (2) *element-coverage entries*—894 non-SC compounds (284 synthetic plus 610 drawn from Materials Project) covering rare elements (Kr, Rn, Fr, Ra, Po, At, and select actinides) that previously had zero or near-zero training examples, eliminating decoder hallucination of never-seen element tokens; and (3) *doping-boundary entries* documenting SC/non-SC transitions within substitution series. The final training CSV contains 60,615 examples (26,178 SC, 34,437 non-SC); the model trains on 60,570 of them (26,133 SC, 34,437 non-SC), withholding the holdout superconductors present in the source data.

The expanded phase-transition corpus (85,186 examples). After the final model was trained, the corpus was expanded again—to 85,186 examples (26,425 SC, 58,761 non-SC)—specifically to supply labels for the phase-transition heads of Sections 4.6 and 5: 18,921 examples carry ferromagnetic Curie temperatures (largely ingested as new examples from the NEMAD magnetic database), 10,039 carry Néel temperatures, 4,701 carry Debye temperatures, and 2,113 carry melting points. These examples post-date the final model: *none* of the encoder/decoder, T_c , or generative results in this paper draw on them. They train the auxiliary heads—the Curie head learns against a frozen encoder

(Section IV)—so the expanded corpus grows what the latent can be asked to read out, without changing the latent whose anatomy this paper reports.

2.2 Data Pipeline

The pipeline consists of three stages. Stage 1 parses 16,521 superconductor entries from NIMS SuperCon, converting decimal formulas to fraction notation (e.g., $\text{YBa}_2\text{Cu}_3\text{O}_{6.9}$ to $\text{Y}(1/1)\text{Ba}(2/1)\text{Cu}(3/1)\text{O}(69/10)$), computing 145 Magpie features, and assigning superconductor family categories. Stage 2 ingests 19,058 raw NEMAD entries, filters to 15,339 experimental-only records, extracts T_c , normalizes formulas through pymatgen, deduplicates against SuperCon by canonical composition, and excludes the generative holdout set. Stage 3 loads non-superconductors from 4 sources (NEMAD magnetic, thermoelectric, anisotropy, and Materials Project), deduplicates across sources, and removes any compositions matching known superconductors or the holdout set.

All deduplication uses canonical pymatgen compositions, making the pipeline robust to formatting differences. Feature schemas are identical across all outputs (151 base columns).

2.3 Why Include Non-Superconductors?

The non-SC materials serve three purposes developed iteratively during training:

Chemical vocabulary expansion. Superconductor formulas cluster around certain elements (Cu, O, Ba, Sr, La, Fe, As) and stoichiometric patterns. Early in training, the formula decoder struggled to reconstruct compositions with rare elements or unusual doping fractions. Non-SC materials from Materials Project and NEMAD expose the encoder and decoder to broader chemical vocabulary. A targeted batch of 3,500 MP entries from rare element neighborhoods (Ti+Li, Hg+Ce, Gd+Pb) was added later in training specifically to address reconstruction failures on holdout-relevant compositions. The NEMAD integration was particularly impactful for underrepresented elements: NEMAD provided 95.1% of all hydrogen-containing SC samples (405 of 426 total) and introduced argon as an entirely new element not present in the original SuperCon dataset.

Class balance. The approximately 43/57 SC/non-SC split (26,133 vs. 34,437 in the final training set) prevents the SC classifier from learning a trivial majority-class prior. Training on all SC with few non-SC teaches “everything is superconducting”; training on all known compounds teaches “nothing is superconducting.”

Phase boundary geometry. Non-SC materials force the latent space to encode the SC/non-SC phase boundary as a geometric feature, providing gradient signal for the SC classifier head.

2.4 Holdout Set

A set of 429 superconductors spanning all major families is reserved entirely from training as the generative holdout: these materials are never presented to the model—not to the encoder, the decoders, or any head—in any form. They are the paper’s central test of generalization. Rather than measuring reconstruction on data the model has already seen, we ask whether latent-space search can *regenerate* each held-out superconductor after the fact, recovering both its composition and its T_c . It can: all 429 targets are recovered under a joint composition-and- T_c criterion (Sections 4.7 and 4.8). The holdout’s construction, provenance, family composition, and full per-candidate recovery are documented in the accompanying supplementary report, *Hydra MAGE—Holdout Set: Construction, Provenance, and Per-Candidate Recovery*.

3 Methods

3.1 Architecture Overview

Hydra MAGE (~120M parameters) encodes chemical formulas into a 2048-dimensional latent vector z and reads it out through a suite of task-specific heads—ten prediction heads plus two complementary

formula decoders, the many-headed (“hydra”) design that names the architecture¹ (a later frozen-encoder Curie head extends the count). Fig. 1 shows the architecture. The parameter count is dominated by the two formula decoders: the autoregressive decoder is $\sim 95.6\text{M}$ —dominated by a 12-layer Transformer (50.4M) and a latent-to-memory projection (42.0M) that maps the 2048-dim latent $\mathbf{z}$ into 16 memory tokens for cross-attention, with the remaining $\sim 3\text{M}$ in embeddings and output projection—while the two-stage decoder adds $\sim 13.9\text{M}$. The three encoder branches together with all prediction heads make up the remaining $\sim 10.3\text{M}$, most of which is the heads rather than the encoder proper (the fusion and μ/σ projection layers are small).

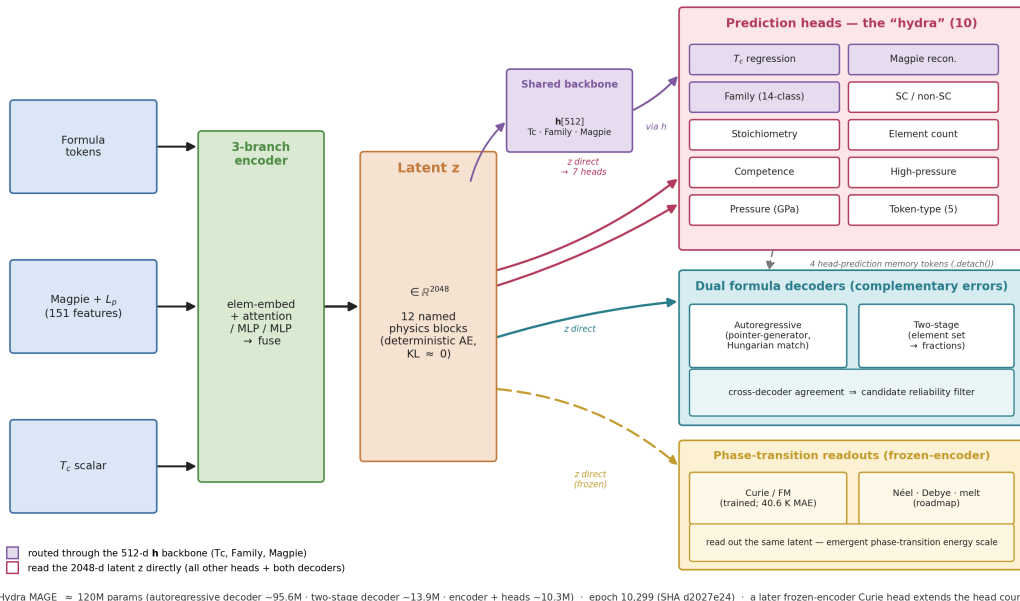


Figure 1: Hydra MAGE architecture. Three encoder branches (element composition, 151 compositional features, T_c) fuse into a deterministic 2048-dimensional latent $\mathbf{z}$ (12 named physics blocks; $\text{KL} \approx 0$), read out by the model’s many heads: ten prediction heads (top), two complementary formula decoders whose cross-agreement filters candidates (middle), and frozen-encoder phase-transition readouts (bottom—the trained Curie head plus the Néel/Debye/melting roadmap). Most heads read $\mathbf{z}$ directly; the T_c , Magpie-reconstruction, and family heads read through a shared 512-d backbone $\mathbf{h}$. Predicted T_c , SC probability, family, and element count are projected—with `.detach()`—into cross-attention memory tokens conditioning the autoregressive decoder. $\sim 120\text{M}$ parameters, dominated by the two formula decoders (autoregressive $\sim 95.6\text{M}$, two-stage $\sim 13.9\text{M}$).

Encoder. A three-branch Full Materials Encoder processes: (1) the tokenized chemical formula through learned element embeddings ($\text{Embedding}(118, 128) \rightarrow 8\text{-head self-attention} \rightarrow 256\text{-dim}$), (2) the 151-dimensional compositional feature vector ($145 \text{ Magpie} + 6 L_p\text{-norm}$) through an MLP ($151 \rightarrow 512 \rightarrow 256$ with GELU), and (3) the scalar T_c value through a conditioning branch ($1 \rightarrow 128 \rightarrow 256$ with GELU). The three 256-dim branch outputs are concatenated (768-dim), fused through $\text{Linear} \rightarrow \text{LayerNorm} \rightarrow \text{GELU}$, and projected to $\mathbf{z} \in \mathbb{R}^{2048}$ via μ and σ heads.

Shared decoder backbone. $\mathbf{z}$ is processed through two shared layers:

$$\begin{aligned} \mathbf{z}[2048] &\rightarrow \text{Linear}(2048, 256) \rightarrow \text{LN} \rightarrow \text{GELU} \\ &\rightarrow \text{Linear}(256, 512) \rightarrow \text{LN} \rightarrow \text{GELU} \rightarrow \mathbf{h}[512] \end{aligned} \quad (1)$$

Only a subset of heads consume this backbone: the T_c head (and its bucket classifier), the Magpie-reconstruction head, the family classifier, and the formula decoder’s attended-input projection branch

¹The “hydra” metaphor—a shared body feeding many task-specific heads—also appears in the machine-learning literature, most directly in the multi-branch HydraNet architecture of Mullanpudi et al. (50) and informally for multi-task perception networks; I use it here for the many-headed readout of a single shared latent.

from **h**. The remaining heads—SC classifier, competence, stoichiometry/fraction, element-count, high-pressure and pressure regressors, and the phase-transition readouts—together with *both* formula decoders read the 2048-d latent **z** directly (the two-stage decoder through its own **z**-projection, the autoregressive decoder through its latent-to-memory projection). The heads that read **z** directly retain full access to the latent; only the **h**-routed heads pass through the 512-d bottleneck.

Decoder heads: (1) *Autoregressive formula decoder*: a 12-layer Transformer attending to 24 cross-attention memory tokens (16 from the latent-to-memory projection plus 8 auxiliary tokens—4 carrying stoichiometry and 4 carrying head predictions), using a pointer-generator with copy mechanism and fraction-aware tokenization, reconstructing formulas token-by-token. Hungarian matching (51) is used for parallel slot-based prediction, eliminating autoregressive ordering constraints during training; (2) *T_c prediction head*: projection through a residual block to normalized $\log(1 + T_c)$ space; (3) *SC/non-SC classifier*: binary classification; (4) *Family classifier*: 14-class hierarchical classification; (5) *Magpie reconstruction head*: reconstructs 151 compositional features; (6) *Stoichiometry head*: predicts element counts and fractions. The final model additionally carries a high-pressure classification head with a paired pressure-magnitude regressor (trained on 226 curated GPa labels), an element-count head, and a competence head predicting reconstruction quality.

Dual decoders with complementary error profiles. Alongside the autoregressive (AR) formula decoder, the final model carries a *two-stage* decoder that first predicts the discrete element set, then predicts the stoichiometric fractions conditioned on that set. The two decoders make different kinds of mistakes: on the holdout, the two-stage decoder achieves higher average roundtrip fidelity than the AR decoder (mean composition similarity 0.981 vs. 0.948) but never produces exact matches (its best similarity is 0.9995, reaching 1.0 on none of the targets), because its continuous fraction regression rarely lands on the precise rational stoichiometry, whereas the AR decoder’s discrete token generation can. Because the two pathways factorize the problem differently—the two-stage decoder separates discrete element identity from continuous stoichiometry explicitly—they fail differently, and that is the point. In a 2048-dimensional latent space explored along many directions at once, bad generations are *expected*: most sampled **Z**-vectors will not decode to coherent materials, and no single decoder can be trusted to know when it has drifted. Two complementary decoders turn this into a self-checking system: when both pathways agree on a candidate, confidence is high; when only one produces a sensible formula, post-processing on the model’s own outputs weeds out the bad generation rather than letting it through. Critically, the two-stage decoder has *no autoregressive failure mode at all*—it predicts the element set and the fractions in single passes, so the token-by-token error compounding that afflicts AR decoding simply cannot occur there. The fact that both pathways operate from the same **z** is additional evidence that the latent representation itself factorizes element identity from stoichiometric detail. The two pathways also respond differently to *conditioning*: when generation is conditioned on a swept target T_c at fixed **z**, the two-stage fraction decoder deterministically shifts the generated doping stoichiometry at fixed elements ($\sim 10\times$ more T_c -responsive), while the autoregressive decoder is largely composition-locked and barely moves. This asymmetry—which is what makes their cross-agreement an informative out-of-distribution signal—is quantified, and its physical fidelity assessed, in Sections 4.11 and 4.13.

Token-type classifier with hard vocabulary masking. A 5-class token-type head (element / integer / fraction / special / end-of-sequence) predicts the type of the next token at every decoding step, and the vocabulary is hard-masked to that type during autoregressive generation. Type confusion (e.g., emitting an element symbol where a fraction numerator belongs) accounted for 30% of AR errors before this head existed—and roughly half during the expanded-vocabulary tokenizer experiments described below; the hard mask eliminates this error class by construction rather than by learning. The head was introduced during those experiments but survived the rollback: it is present and its loss active in the final model, and the production generation path applies the hard mask at every step. The decoder memory is additionally enriched with the projected outputs of four prediction heads, conditioning generation on the predicted material type.

T_c as decoder conditioning signal. The predicted T_c , SC probability, family classification, and element count are projected into 4 cross-attention memory tokens (52) ($24 \rightarrow 256 \rightarrow 512 \rightarrow 4 \times 512$ dims) that the formula decoder attends to at every generation step. During training, these predictions are passed with `.detach()`, preventing gradient backpropagation through the T_c head from the decoder but ensuring the shared backbone **h** must serve both tasks simultaneously. This creates a dual supervision mechanism: the T_c head is trained by its own regression loss, and it must also produce conditioning signals that help the formula decoder—if the T_c prediction is poor, the formula decoder

loses a useful attention signal. Importantly, the T_c head is only active for materials classified as superconducting; when the SC classifier predicts non-SC, the T_c head is bypassed and a fill-in value of 0 K is assigned. This means that a reported T_c of 0.0 K in the evaluation results indicates an SC classification failure, not a T_c prediction of zero.

3.2 The 2048-Parameter Effective Theory

The latent space dimensionality is a deliberate design choice driven by the incompleteness of the input data. A typical superconductor formula is specified by 8–12 numbers (element identities and stoichiometries); the Magpie representation expands this to 145 features. The encoder maps every material into 2048 dimensions—roughly $200\times$ larger than the raw compositional specification.

This expansion provides representational capacity for *hidden variable inference*. The encoder cannot observe crystal structure, oxygen ordering, or apical oxygen distances, but if certain compositions consistently appear with certain structural motifs, the encoder can learn statistical proxies from compositional patterns. Importantly, selectively including structural data where available would introduce systematic training bias, since structural information is well-characterized for some families (BCS metals, some cuprates) but sparse for others (organics, heavy-fermion, many doped variants). Using composition uniformly across all families avoids this inhomogeneity.

The 2048 dimensions are partitioned into 12 named coordinate blocks (Table 2). Blocks 1–2 (Ginzburg-Landau, BCS) are constrained by physics self-consistency losses. Block 8 (compositional) is actively supervised against 15 formula-derivable targets. Block 11 (Magpie encoding) learns a linear projection of the 145 Magpie features into 62 latent dimensions. Block 12 (dims 512–2047) is entirely unsupervised discovery space.

Table 2: Physics Z coordinate blocks

Block	Name	Dims	Status
1	Ginzburg-Landau	0–19	Self-consistency
2	BCS/Microscopic	20–49	Self-consistency
3	Eliashberg	50–69	Placeholder
4	Unconventional	70–109	Placeholder
5	Structural	110–159	Placeholder
6	Electronic	160–209	Placeholder
7	Thermodynamic	210–269	Placeholder
8	Compositional	270–339	Active
9	Cobordism	340–399	Self-consistency
10	Dim. Ratios	400–449	Self-consistency
11	Magpie Encoding	450–511	Active
12	Discovery	512–2047	Unsupervised

The decoder heads create self-consistency learning channels: the family classifier forces $\mathbf{z}$ to encode family-discriminating features; the SC classifier forces the phase boundary; the Magpie head preserves compositional statistics. The T_c head operates on a latent space already shaped by these competing constraints. I call the resulting representation a *2048-parameter effective theory of superconductivity*—compare with BCS/Allen-Dynes (53) (3 parameters: λ , $\omega_{\log}$, μ^* ; applicable to one family) or the Presland dome (54) (3 parameters: $T_{c,\max}$, p_{opt} , dome width; applicable to one cuprate sub-family).

3.3 Training

The model is trained with Adam ($\text{lr} = 10^{-4}$, batch size 1024) on a single NVIDIA A100 80GB GPU via Google Colab, implemented in PyTorch (55). Despite the historical “VAE” nomenclature in the codebase, KL divergence is weighted at 10^{-4} —effectively zero—making the model a *deterministic autoencoder* rather than a variational model. This was a deliberate choice: the generative capability comes from the autoregressive formula decoder, not from sampling in the latent space, and the near-zero KL weight allows the encoder to organize $\mathbf{z}$ freely without being pulled toward a unit Gaussian.

The training objective comprises 15 active loss terms organized in two layers; Table 3 summarizes the full suite. The core losses computed jointly include: formula cross-entropy with focal weighting (56) ($\gamma = 2.0$, label smoothing 0.05), T_c regression (Huber loss with asymmetric underprediction penalty, Kelvin-weighted to upweight rare high- T_c samples), Magpie reconstruction, stoichiometry prediction, element count prediction, constraint zoo penalties (site occupancy, charge balance), and L2 regularization. Auxiliary losses computed externally include: SC/non-SC classification, family classification (14 classes), high-pressure classification (a dedicated binary head trained specifically for JARVIS hydrides, distinct from the family classifier), stop prediction (dedicated end-of-sequence head), and token type classification (element/integer/fraction/special—used at inference to hard-mask invalid token types).

Table 3: The full loss suite. Groups: core losses (computed jointly), auxiliary head losses, physics self-consistency losses on latent blocks, sequence-level reinforcement, and the separately-staged frozen-encoder pretraining losses. Weights evolve under the curriculum/skip scheduler described below.

Loss	Target	Form / purpose
<i>Core (joint)</i>		
Formula reconstruction	AR decoder	Cross-entropy, focal $\gamma=2.0$, label smoothing 0.05
T_c regression	T_c head	Huber on normalized $\log(1+T_c)$; asymmetric underprediction penalty; Kelvin-weighted
Magpie reconstruction	Magpie head	Regression on 151 compositional features
Stoichiometry	fraction head	Element molar fractions
Element count	count head	Number of distinct elements
Constraint zoo	decoder outputs	Site-occupancy and charge-balance penalties
L2 regularization	all weights	Standard weight decay
<i>Auxiliary heads</i>		
SC/non-SC	SC head	Binary cross-entropy
Family	family head	14-class hierarchical cross-entropy
High-pressure	HP head	Binary (hydride-focused)
Pressure magnitude	pressure head	Huber on $\log(1+\text{GPa})$, 226 curated labels
Stop prediction	EOS head	Dedicated end-of-sequence supervision
Token type	type head	5-class cross-entropy; hard vocab mask at inference
<i>Physics self-consistency (latent blocks; no external data)</i>		
Ginzburg–Landau	$\mathbf{z}[0-19]$	$\kappa = \lambda_L/\xi$, $H_{c2} \propto 1/\xi^2$, H_{c1} relation
BCS	$\mathbf{z}[20-49]$	$\xi_0 \propto v_F/\Delta_0$; gap-ratio soft bounds
Cobordism / dim. ratios	$\mathbf{z}$ blocks 9–10	Dimensionless-ratio consistency
<i>Sequence-level RL and prior</i>		
SCST	AR decoder	Greedy-baseline sequence reward with constraint-aware penalties
KL divergence	$\mathbf{z}$	Weight 10^{-4} (effectively zero $\rightarrow$ deterministic autoencoder)
<i>Separately staged (frozen encoder; Section on phase-transition heads)</i>		
FM classification	FM head	Binary cross-entropy on 18,921 Curie-labeled examples
Curie regression	Curie head	Huber ($\delta=1.0$), underprediction penalty 1.5

3.3.1 Closing the Teacher-Forcing Gap with SCST

To close the gap between teacher-forced training and autoregressive inference, I employ Self-Critical Sequence Training (SCST) (57). This addresses a critical training challenge: with teacher forcing (feeding ground-truth tokens at each step), the model reached 99% exact formula reconstruction, but when generating autoregressively (feeding its own predictions), accuracy was initially as low as 17%. This 82-point gap occurs because small per-token errors compound during autoregressive generation, causing the model to drift into unfamiliar token sequences.

My initial attempt to close this gap used REINFORCE with Leave-One-Out (RLOO) baseline, but I discovered a critical implementation bug: averaging the RLOO advantages across K samples before multiplying with log-probabilities mathematically cancels to zero gradient. The model received *no*

RL training signal for 339 epochs (2015–2354), consuming 84% of epoch time (119s/epoch) for zero gradient. The fix was to replace RLOO with SCST (57), which uses the greedy decode reward as baseline with a single sample (2 forward passes vs. RLOO’s 4).

Through SCST, constraint-aware rewards, and iterative architecture refinement, the epoch-3292 checkpoint achieved 86.5% autoregressive exact-match reconstruction, *surpassing* the teacher-forced rate of 85.4% at the same checkpoint—the gap is not just closed but inverted, with the model generating better formulas autoregressively than with teacher forcing. This inversion is likely because SCST’s sequence-level reward directly optimizes whole-formula quality, while teacher forcing optimizes per-token accuracy.

Notably, a subsequent series of tokenizer experiments (semantic fraction tokenization expanding the vocabulary from 148 to 4,355 tokens; isotope tokens; curriculum AR warmup) drove AR accuracy back down to 5–7.5% while TF rose to 95.6% (TF only reached 98.6% after the tokenizer was rolled back). The semantic tokenization traded sequence length for vocabulary precision—reducing sequences by $\sim 50\%$ —but the model failed to learn clean token type boundaries, with type confusion worsening from 30% to 52% of errors. This motivated rolling the tokenizer back to the original 148-token vocabulary and restarting training from there.

SCST reward includes physics-grounded constraint penalties (duplicate elements, impossible element combinations, family-specific stoichiometry violations) that shape generation toward chemically valid formulas.

3.3.2 Physics Z Losses

The physics self-consistency losses for blocks 1–2 (GL, BCS), 9 (cobordism), and 10 (dimensionless ratios) enforce known physical relationships (8; 7; 58) within the latent space without requiring external data. For example, the GL block enforces $\kappa = \lambda_L/\xi$ (8), $H_{c2} \propto 1/\xi^2$ (58), and $H_{c1} \propto (1/\lambda^2)\ln(\kappa)$; the BCS block enforces $\xi_0 \propto v_F/\Delta_0$ (7) and soft-bounds the gap ratio $2\Delta_0/(k_B T_c) \in [1, 5]$. These losses were activated during training, converged, and were subsequently deprioritized by the smart loss scheduler as the primary challenge shifted to closing the teacher-forcing/autoregressive gap.

3.3.3 Curriculum and Scheduling

Loss weights evolve during training via curriculum scheduling (T_c and Magpie ramp up over 30 epochs) and smart loss skipping (converged losses computed every 4th epoch, resuming on regression).

3.3.4 Training Progression

Table 4 shows the evolution of key metrics across training stages. The epoch-3292 checkpoint marked the point where AR accuracy first surpassed TF; the expanded-vocabulary tokenizer experiments that followed degraded AR performance and were rolled back. Training then continued on the restored tokenizer—with a net2net-widened (function-preserving widening) T_c head, a restored SC head, the hierarchical family head, an element-count head, and the two-stage decoder added along the way—for roughly 7,000 further epochs to epoch 10,299. That checkpoint (SHA d2027e24) is the **final model**—hereafter simply “the model.” Every result in this paper is reported against it—the one exception, the validity-head successor of Section 4.8, is explicitly identified wherever it appears; the few comparisons to the epoch-3292 checkpoint are explicitly labeled as historical, and the two are genuinely different weights (the T_c head, among others, was widened and then trained $\sim 7,000$ further epochs).

Two architectural experiments from this period are instructive failures: (1) *Semantic fraction tokenization*—collapsing each fraction like (17/20) from 7 tokens to 1—shortened sequences but, as noted above, worsened type confusion rather than helping, and was rolled back. (2) *Skip connections* from encoder to decoder memory tokens were removed after discovering they bypassed the information bottleneck: TF exact was 80.6% but AR exact was 0.1%, indicating the decoder relied entirely on leaked encoder information during teacher forcing.

In the final model, the strongest error predictors for greedy decoding over all 60,570 training examples remain the compositional-complexity variables—element count ($r = 0.159$ with token-error count)

Table 4: Training progression and checkpoint lineage. TF Exact is teacher-forced exact-match; AR Exact is single-shot greedy autoregressive true-exact-match on training data. The semantic-fraction and isotope/type-head experiments (rows 4–5) expanded the vocabulary but collapsed AR decoding and were rolled back; single-shot AR fidelity falls from its epoch-3292 peak and does not fully recover, but ceases to be the operative metric once generation moves to Z-space search with cross-decoder filtering (Section 4.7).

Training stage	Epoch	TF Exact	AR Exact	Vocab
Early	1778	73.9%	49.8%	148
Plateau	2522	59.8%	~50%	148
SCST-optimized	3292	85.4%	86.5%	148
Semantic fractions (exp.)	3475+	95.6%	5–7%	4,355
+ isotopes, type head (exp.)	4000+	95.6%	7.5%	4,647
Tokenizer rollback	4330+	98.6%	14–17%	148
Final model	10,299	98.6%	22.0%	148

and sequence length ($r = 0.124$)—while T_c ($r = -0.018$) and latent norm ($r = -0.028$) carry essentially no signal. The ordering matches the epoch-3292 analysis (where the corresponding correlations were 0.55–0.59 under a different error instrument), but the absolute correlations are far weaker: decode errors in the final model are much less predictable from surface features.

Which latent blocks predict decode success. Comparing per-block latent norms between samples the greedy decode reproduces exactly and those it misses: at the final model the *Discovery* block (the unsupervised 75% of $\mathbf{z}$) shows the largest standardized gap (Cohen’s $d = 0.64$, larger norms on correctly-decoded samples), followed by the *Electronic* ($d = 0.42$), *Magpie* ($d = 0.40$), *Structural*, *GL*, and *Eliashberg* blocks ($d \approx 0.37$). Notably, at epoch 3292 the *Compositional* block was the strongest discriminator; in the final model it is nearly neutral ($d = -0.11$)—over extended training, the information that determines decodability appears to have migrated from the supervised compositional block into the unsupervised *Discovery* space.

Why weak single-shot AR decoding is not a bottleneck. The final model has a property that looks alarming out of context: teacher-forced accuracy on the reverted line is near-perfect (98.6%), yet a single greedy autoregressive decode reproduces only 22.0% of training formulas exactly (44.4% of superconductors). The remarkable empirical fact is that this AR gap does not propagate to the metric that matters. Generation never relies on one decode being right: the Z-space search samples the latent space densely ($\sim 27,000$ unique formulas per target, Section 4.7), and the model’s own heads—SC probability, T_c , family, *Magpie* consistency, plus cross-decoder agreement between the AR and two-stage pathways—filter the candidates. Under that protocol the same checkpoint recovers held-out superconductors that single-shot AR decoding misses (Section 4.7). Single-shot AR fidelity turns out to be the wrong quantity to optimize for a generative-search system: what matters is that correct formulas exist at reachable latent locations and that the model can recognize them when sampled—and the two-stage decoder, which has no autoregressive drift by construction, provides a decoding pathway for which the gap never existed.

3.3.5 Phase-Transition Heads (Curie MVP)

Motivated by the emergent phase-transition finding of Section 4.6, a ferromagnetic-Curie-temperature head (paired binary FM classifier + gated Curie regressor, mirroring the SC/T_c head design) was added on the shared backbone. When a pre-flight audit found the initially-curated labels ($\sim 2,600$ FM positives matched into the existing corpus) insufficient, the corpus itself was expanded: unmatched NEMAD magnetic materials were ingested as new examples with full *Magpie* features, bringing Curie-labeled examples to 18,921 within the 85,186-example expanded corpus of Section II. Training is deliberately two-stage: a pretraining stage *freezes the encoder and every existing head* and trains only the FM and Curie heads—so the head learns to *read out* what the shared latent already contains rather than reshaping it—before any integrated fine-tuning. Results are reported in Section 4.6; Néel, Debye-temperature, and quantile-uncertainty heads are designed but not yet trained.

4 Results

4.1 T_c Prediction Performance

The T_c head achieves $R^2 = 0.9999$ with $\text{MAE} = 0.10$ K on all 60,570 training examples (0.22 K on the 26,133 superconductors), with near-perfect reconstruction across all 14 family classes and the full T_c range (0–300 K). The residual structure is equally clean: mean signed residual on superconductors is +0.098 K, and across all 34,437 non-superconductors (true $T_c = 0$) the maximum falsely predicted T_c is 0.031 K—the model never hallucinates a transition temperature for a non-superconductor.

To test whether this performance reflects learned physics or compositional shortcuts, I train a Magpie-only baseline MLP (Magpie[145] $\rightarrow 256 \rightarrow 128 \rightarrow 1$) with the same compositional information but no latent space structure. The baseline achieves $R^2 = 0.920$, $\text{MAE} = 3.22$ K—roughly $30\times$ the final model’s error. (The baseline was trained on the earlier 52,813-sample dataset; its performance is consistent with the composition-only ceiling across the field, Table 5.)

Table 5: Composition-based T_c prediction benchmarks

Method	Year	R^2	MAE (K)	Ref
Random Forest + Magpie	2018	0.88 [†]	N/A	(12)
XGBoost	2018	0.92 [†]	9.5 [‡]	(11)
SOAP + ML (structural)	2022	0.929 [†]	N/A	(22)
GBFS	2024	0.952 [†]	3.73 [†]	(14)
Magpie baseline	2026	0.920*	3.22*	–
Hydra MAGE (ep. 10299)	2026	0.9999*	0.10*	–

[†]Test-set metrics. *Training-set metrics (appropriate for interpretability analysis; baseline provides controlled ablation). [‡]RMSE (MAE not reported). N/A indicates MAE not reported in cited work. The $\sim 30\times$ error reduction ($\text{MAE } 3.22 \text{ K} \rightarrow 0.10 \text{ K}$) quantifies the contribution of the multi-task latent space beyond raw composition.

The $\sim 30\times$ error reduction quantifies the contribution of the multi-task latent space beyond raw composition. If the T_c head merely learned compositional shortcuts (e.g., “materials with Cu and O have high T_c ”), it could not outperform a direct Magpie $\rightarrow T_c$ MLP by this margin.

4.2 Latent Space Analysis

UMAP visualization. Fig. 2 shows UMAP (59) embeddings of the latent vectors $\mathbf{z}[2048]$ for all 60,570 training examples in the final model. In the left panel (colored by family), the non-SC materials (light blue) form a large dispersed multi-lobed region that is well-separated from the superconducting families. The cuprate sub-families occupy a distinct high- T_c archipelago with internally structured sub-clusters (LSCO, Cuprate-Other, BSCCO, TBCCO, HBCCO each concentrated in their own territory), and the iron-based and BCS families form separate islands. The right panel (colored by $\log(1 + T_c)$) reveals continuous temperature gradients within the superconducting islands—the cuprate region glows green-to-yellow (highest T_c) while the non-SC region sits uniformly at the bottom of the color scale—confirming that the latent geometry encodes both family identity and T_c simultaneously.

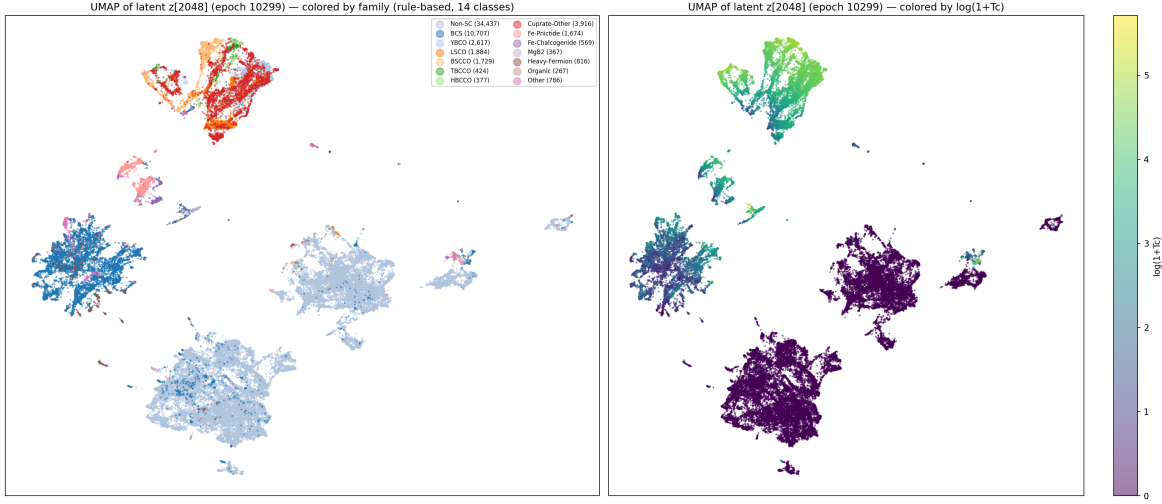


Figure 2: UMAP of the latent space $\mathbf{z}[2048]$ for all 60,570 training examples in the final model (epoch 10,299). Left: colored by superconductor family (rule-based, 14 classes). Right: colored by $\log(1 + T_c)$. Non-SC materials (light blue) form a large dispersed region well-separated from the SC families; the cuprate archipelago carries the highest- T_c gradient.

t-SNE visualization. Fig. 3 shows t-SNE (60) embeddings of the same latent vectors. The same structure appears under a different projection: a large dispersed non-SC cloud, tightly clustered cuprate sub-families carrying the highest T_c values, and separate BCS and iron-based territories with continuous T_c gradients inside each cluster.

This structure demonstrates that the latent space has self-organized into a physically meaningful manifold where both family identity and T_c are geometrically encoded—the model represents the diverse mechanisms of superconductivity in a single continuous space.

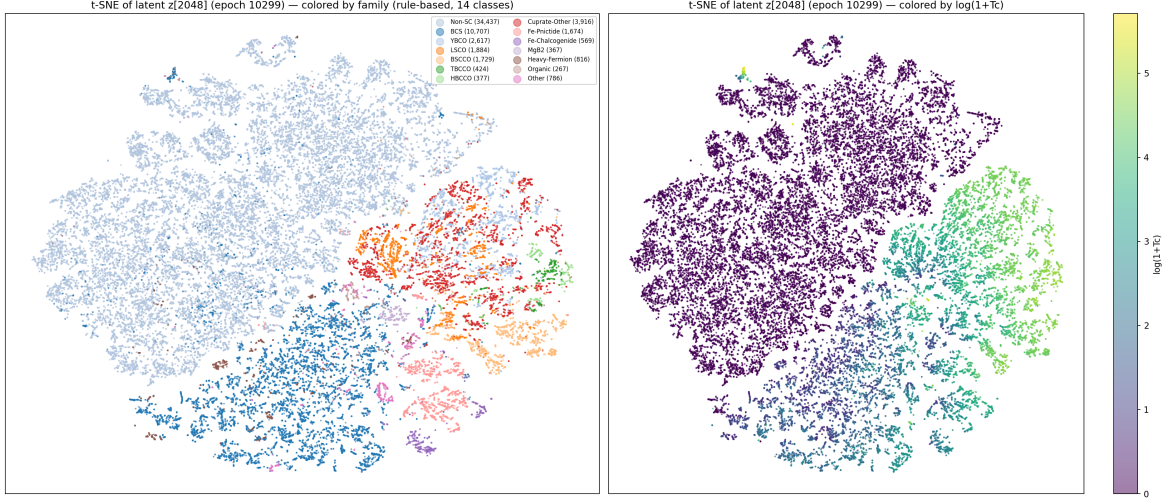


Figure 3: t-SNE of the latent space $\mathbf{z}[2048]$ for all 60,570 training examples in the final model. Left: colored by superconductor family (14 classes). Right: colored by $\log(1 + T_c)$. Non-SC materials form a distinct dispersed region; cuprate sub-families cluster tightly with continuous T_c gradients.

4.3 Per-Family Gradient Analysis

Gradient sensitivity $|\partial T_c / \partial z_d|$ reveals which latent dimensions the T_c head uses for each material family. Fig. 4 shows hierarchical clustering of families by their mean gradient profiles in the final model (400 samples per family, cosine distance, average linkage).

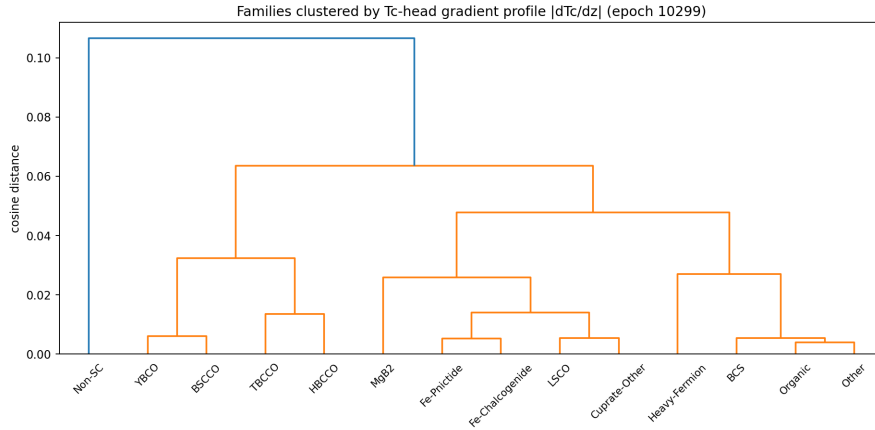


Figure 4: Hierarchical clustering of superconductor families by T_c gradient profile at epoch 10,299 (cosine distance, average linkage). Non-SC is maximally isolated. Multi-layer cuprates (YBCO, BiSCCO, TBCCO, HfBCCO) form a distinct sub-branch; LSCO clusters with the iron families rather than with the multi-layer cuprates.

Key findings: (1) Non-SC is maximally isolated (cosine distance ~ 0.11), confirming the model has learned a genuine SC/non-SC distinction in its T_c computation, not just in the classifier head. (2) Multi-layer cuprates (YBCO, BSCCO, TBCCO, HBCCO) form a distinct sub-branch, separate from every other mechanism class. (3) LSCO clusters with the iron-based families and Cuprate-Other rather than with the multi-layer cuprates, echoing the shared quasi-2D, single-layer physics observed at the earlier checkpoint. (4) Compared with the epoch-3292 dendrogram, MgB_2 is no longer the second-most-isolated family: after $\sim 7,000$ further epochs it has moved to the base of the iron-family branch, while the overall Non-SC-first, multi-layer-cuprates-together topology is stable—family-level gradient structure persists across a $3\times$ change in training duration. These family-specific gradient profiles demonstrate that the 2048-parameter effective theory contains distinct sub-theories for different material classes rather than a single universal mapping.

4.4 Extracting the Effective Theory as Equations

If the latent space is a 2048-parameter effective theory, it should be possible to read parts of that theory back out as explicit mathematics. This section distills the T_c head into closed-form per-family expressions via symbolic regression, including one retraction that illustrates why this kind of extraction must be audited.

Method. For each family, gradients $\partial T_c / \partial \mathbf{z}$ are computed over the family’s training examples and reduced by PCA to a small set of principal directions; each sample’s PC scores s_i (normalized to unit variance) become the input variables. PySR (39) (200 iterations, 30 populations, operators $+$, $-$, $\times$, $\div$, square, sqrt, exp, log; nested-exponential constraints; physics-informed parsimony penalties) then fits the *network’s* T_c prediction—not experimental T_c —so the extracted expressions describe what the network computes, with measurement noise excluded by construction.

A retraction, then a correction. The first extraction (at the epoch-3292 checkpoint) produced an exponential form for BCS materials, $y = -1.53 e^{-0.445s_1}$, initially celebrated as an Allen-Dynes-like (53) signature of electron-phonon coupling. Diagnosis showed the analysis sets were contaminated: the per-family sample lists included non-superconducting materials—for BCS.Conventional roughly 70% of samples—and the T_c head’s gradients at non-SC latent vectors are untrained noise, since the head only ever received gradient signal on superconductors. The celebrated exponential was tracing the SC/non-SC *boundary*, not phonon coupling, and was retracted. The entire pipeline was rerun on SC-filtered data in the final model. Notably, exponential structure *reappears* in the corrected BCS and MgB_2 expressions (Table 6)—but now derived from clean, superconductor-only data, where an Allen-Dynes-like reading is at least admissible. The retraction did not kill the finding; it moved it onto defensible ground.

Table 6: PySR-extracted per-family T_c expressions (SC-filtered, final model), all 13 SC families (the 14th family class, non-SC, is excluded from this SC-filtered analysis). R^2 is computed against the network’s T_c predictions over the *full* family data (normalized space); MAE in Kelvin. Gap = train—test R^2 on an 80/20 random split of the fixed expression.

Family	N_{SC}	R^2	MAE (K)	Gap	Form
MgB_2	367	0.984	1.5	−0.007	exponential (BCS-like)
Iron Pnictide	1,647	0.958	1.9	+0.026	exponential
Iron Chalcogenide	568	0.936	2.1	+0.014	linear
Organic	267	0.896	7.9	−0.044	quadratic
Cuprate TBCCO	422	0.871	7.3	+0.006	exponential
Other/Unknown	780	0.846	2.0	−0.017	quadratic
Cuprate HBCCO	376	0.841	8.7	+0.083	exp.–quadratic
Heavy Fermion	810	0.839	1.6	−0.059	exponential
Cuprate Other	3,901	0.793	8.9	+0.032	quadratic
Cuprate LSCO	1,859	0.787	6.5	+0.013	linear
BCS Conventional	10,647	0.772	3.4	+0.008	exponential (BCS-like)
Cuprate BSCCO	1,719	0.692	18.7	−0.047	quadratic
Cuprate YBCO	2,592	0.650	9.2	−0.009	exponential

Two representative expressions (PC scores s_i normalized; output in normalized $\log(1 + T_c)$ space):

$$\text{BCS: } y = -0.344 (-s_3 + e^{-s_1 - 0.643} + e^{-s_2}) \quad (2)$$

$$\text{LSCO: } y = 0.338 (s_2 + 1.364) - 0.140 (s_3 s_5 + s_4 (s_5 + s_7)) \quad (3)$$

With the three smallest cuprate families now rerun as well (Table 6), a class-level pattern is visible: every cuprate family’s expression takes a quadratic (inverted-parabola) or exponential form except single-layer LSCO (linear)—a functional-form pattern across the cuprate class (we do *not* claim it reproduces the physical doping dome; see the caveat below), alongside BCS-like exponentials for the two phonon-mediated families.

Validation and caveats. An 80/20 holdout evaluation of the final expressions shows train–test R^2 gaps within ± 0.06 for 12 of 13 families (maximum $+0.083$, HBCCO)—no material overfitting, consistent with the epoch-3292 extraction where the same instrument showed gaps below 0.04 and PySR’s parsimony pressure prevented the memorization that sank a 325-parameter quadratic fit on TBCCO (test $R^2 = 0.33$). The parsimonious expressions are lossy summaries, not perfect surrogates: R^2 against the network’s own predictions ranges from 0.98 (MgB₂) down to 0.65 (YBCO)—the network’s multi-layer cuprate T_c function occupies 44–52 latent directions (Section 4.5), and an 8-variable closed form cannot capture all of it. Quadratic fits in the same PC coordinates reach $R^2 \geq 0.97$ for all 13 families, confirming the function is low-order in the right coordinates; the PySR forms trade fit for interpretability.

Physics-template fits. A complementary extraction imposes the physics up front instead of discovering the form: fit the *Allen-Dynes formula* (53) to the network’s BCS T_c function with λ , $\omega_{\log}$, and μ^* each parameterized as bounded readouts of the top gradient-PCs, and the *Presland-form template* (54) $T_c = T_{c,\max} [1 - 82.6 (p - 0.16)^2]$ to each cuprate family with $T_{c,\max}(\mathbf{z})$ and a free effective-doping readout $p(\mathbf{z})$. On SC-filtered data these constrained templates fit well—better at matched dimensionality than the free-form PySR expressions: Allen-Dynes reaches 5.7 K RMSE on BCS at $k = 10$ PCs (family T_c std 20.8 K), and the Presland-form parabola reaches 3.3–5.1 K RMSE across *all six* cuprate families at $k = 10$ (family stds 22–30 K), improving steeply with k (e.g., BSCCO: 18.8 K at $k=3 \rightarrow 3.3$ K at $k=10$). At matched dimensionality the parabolic template therefore fits the network’s cuprate T_c function to within ~ 4 K—but this reflects the flexibility of a low-order fit, *not* a physical claim. Here $p(\mathbf{z})$ is a **free fitted readout**, not a measured doping level, and a bounded parabola with two latent-dependent parameters fits many smooth functions; the fit does *not* establish that the network represents hole doping or reproduces the physical doping dome. Direct attempts to navigate a doping axis in the latent (steering and interpolation along Sr stoichiometry) fail to recover a smooth dome, consistent with the model’s dominant fine-stoichiometry error mode. We therefore claim functional-form flexibility only; tying $p(\mathbf{z})$ to real hole concentrations (the linear-probe program of Section 5) is what a physical claim would require, and we make none. Two caveats are first-class results, not footnotes. First, direction-type classification labels *all* gradient PCs as “switching” directions (low Jaccard overlap between formula sets at the 5th vs. 95th score percentiles), meaning the expressions may describe interpolation between discrete material sub-types rather than continuous doping/coupling physics—exactly the failure mode Vafa et al. (40) demonstrated for planetary-dynamics transformers. Second, the unsupervised Discovery block (75% of $\mathbf{z}$) dominates every PC’s norm, confounding block-level attribution. These expressions are therefore offered as audited descriptions of the network’s learned function; whether they are also descriptions of nature requires linear probes tying PC scores to measurable physical variables.

4.5 Anatomy of the Latent Space

This section asks the model-internals question directly: which of the 2048 dimensions does each computation actually use? Four gradient-based censuses—per-family dimensionality, per-head subspaces, dead dimensions, and element encoding—map the answer. All artifacts are keyed to the final model.

Per-family effective dimensionality. Jacobian PCA of $\partial T_c / \partial \mathbf{z}$ shows the T_c computation is dramatically lower-dimensional than the 2048-dim ambient space, and the dimensionality is family-dependent (Table 7, Fig. 5). BCS materials collapse to 3 principal directions at 90% variance (Heavy-Fermion to 2), while cuprates need 7–10 at 90% and 44–52 at 99%. The model spends representation capacity where the physics is hardest: cuprate T_c , with its doping-dependent, multi-layer physics, occupies a far richer subspace than phonon-mediated BCS T_c .

Table 7: Effective dimensionality of the T_c computation per family: number of gradient-PCA components to reach 90% / 99% explained variance (final model). N is the per-family SC-filtered sample count; it differs marginally from the counts in the PySR expression table above, reflecting a slightly different SC-filter snapshot.

Family	N	$k_{90\%}$	$k_{99\%}$
Heavy Fermion	816	2	25
BCS Conventional	10,707	3	33
Iron Chalcogenide	569	3	20
Organic	267	3	14
Other/Unknown	786	3	22
Iron Pnictide	1,674	4	24
MgB ₂	367	4	19
Cuprate LSCO	1,884	4	33
Cuprate Other	3,916	6	47
Cuprate YBCO	2,617	7	52
Cuprate BSCCO	1,729	7	49
Cuprate HBCCO	377	8	44
Cuprate TBCCO	424	10	51

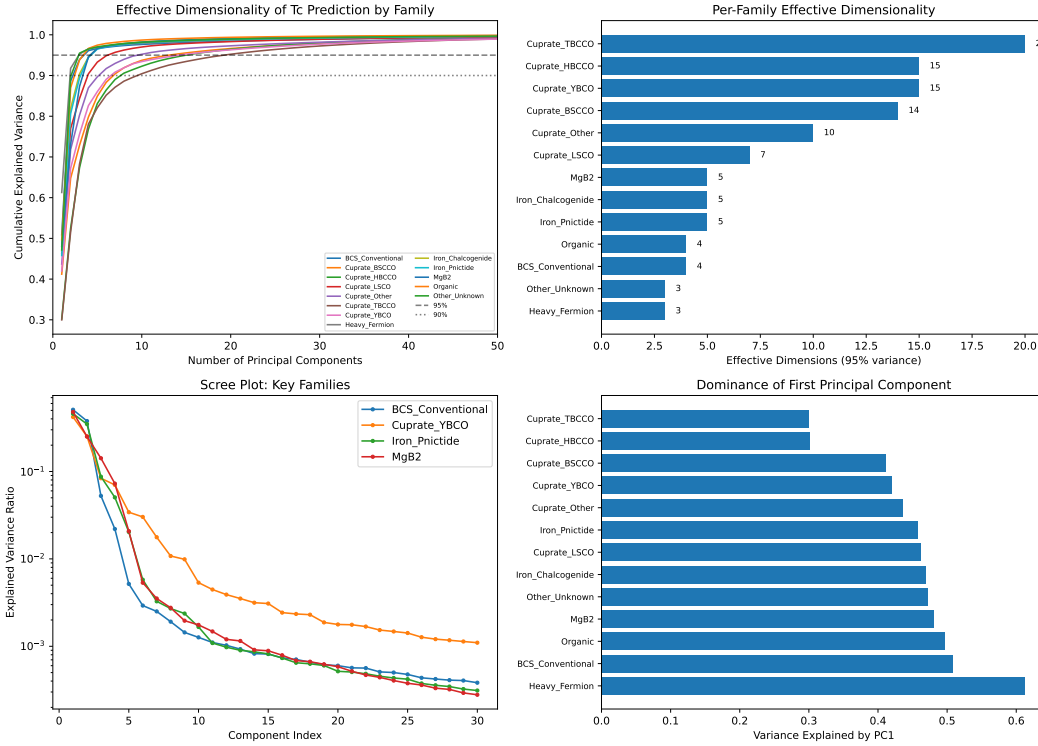


Figure 5: Jacobian-PCA summary in the final model: per-family cumulative-variance scree curves, effective dimensionality at 99% variance, decay of component variance, and first-PC dominance. Cuprate families require substantially more principal directions than BCS/iron families.

Per-head subspaces. Extending the same census to all prediction heads reveals a striking division of labor. The T_c head operates in 14–52 directions (at $R^2 \geq 0.99$) depending on family; the SC classification head spans 74–100 directions—its sensitivity is distributed across the whole space because it consumes every other head’s prediction through the cross-head consistency chain; the high-pressure head uses ~ 34 –54; the family head varies from 7 (TBCCO) to 46 (BCS). The competence head is the extreme case: for *every* family, a single principal direction explains $\geq 99.4\%$ of its gradient variance—the network computes “will I reconstruct this material correctly?” along literally one axis of the 2048-dimensional space.

Dead-dimension census. With thresholds $\epsilon_{\text{abs}} = 10^{-4}$, $\epsilon_{\text{rel}} = 0.01$ on per-head gradient magnitudes, **zero** of the 2048 dimensions are dead across all heads: every dimension carries signal for at least one computation. Head-specific vestigial dimensions exist—the competence and family heads each ignore 45 dimensions—but the T_c , SC, and HP heads use the full space.

How elements are written into $\mathbf{z}$. A $[2048 \times 118]$ Pearson correlation matrix between latent dimensions and element molar fractions shows element identity is encoded redundantly and distributedly: 2,045 of 2,048 dimensions (99.9%) carry element signal at $|r| \geq 0.10$, and 56.6% of all (dimension, element) pairs are significant after Bonferroni correction ($\alpha = 2.1 \times 10^{-7}$). Yet individual elements also have remarkably concentrated encodings: oxygen correlates at $r = +0.83$ with a single dimension (276), hydrogen at $r = +0.81$ with dimension 272, and copper has a *bipolar* encoding— $r = +0.70$ on dimension 888 and $r = -0.70$ on dimension 436—suggesting a learned contrast between Cu-containing material classes. These correlation vectors are directly actionable: Section 4.9 uses them to push generated candidates toward or away from specific elements.

Calibration landscape. HDBSCAN (61) clustering of the T_c error landscape over all 26,133 superconductors finds no pathological regions: global mean signed residual +0.098 K, mean $|\text{residual}|$ 0.217 K, zero clusters with $|\text{mean residual}| > 5$ K, and—as reported in Section 5.1—zero T_c hallucinations above 1 K across all 34,437 non-superconductors. For generation this means there are no known biased $\mathbf{z}$ -regions that candidates must avoid.

Drift across training. Comparing these censuses against the same analyses at epoch 3292 quantifies how the internal theory evolved over $\sim 7,000$ epochs: cuprate effective dimensionality broadened dramatically ($k_{99\%}$: YBCO 27 $\rightarrow$ 52, HBCCO 23 $\rightarrow$ 44, TBCCO 34 $\rightarrow$ 51) while BCS, iron-based, MgB_2 , and organic families stayed essentially fixed. The model refined its hardest sub-theory—cuprate T_c —across more latent directions while leaving converged sub-theories untouched.

4.6 An Emergent Phase-Transition Energy Scale

The most surprising internal finding was not planned; it emerged from auditing the model’s mistakes.

The discovery. During a training-data audit, a set of materials labeled as superconductors (with recorded “ T_c ” values) turned out to be magnetic materials whose phase-transition temperatures had been mis-ingested from source databases. The model’s predictions on these mislabeled materials were not noise: they matched the materials’ *real* (non-superconducting) transition temperatures. EuO: predicted 69.0 K vs. 69.0 K Curie temperature. EuS: 16.5 K vs. 16.5 K (88). A series of $\text{Ga}_{1-x}\text{Mn}_x\text{As}$ dilute magnetic semiconductors: 80, 100, 110, 140 K (89), each exactly matching its Curie temperature. Overall, 9 of 11 audited cases fell within 30 K of the material’s true magnetic or structural transition temperature, 6 of them exactly.

Ruling out memorization. The exact matches above could reflect leaked labels (the mislabeled entries were, after all, in training). The critical test is materials *not present* in the pre-audit training data: on five such compounds ($\text{Co}_3\text{Sn}_2\text{S}_2$, FeCr_2S_4 , Mn_3CuN , CrVO_4 , Fe_3GeTe_2), the model’s “ T_c ” outputs match the true Curie/Néel temperatures with MAE 10.6 K, Pearson $r = 0.989$ ($p = 0.0013$), and perfect rank order (Spearman $\rho = 1.0$); a Monte Carlo control finds zero hits this good in 500,000 draws from a uniform null. $\text{Co}_3\text{Sn}_2\text{S}_2$ —a Weyl-semimetal ferromagnet never seen in training—is predicted at 178 K against a measured Curie temperature of 177 K (90).

The interpretation that survives this test: the latent space does not encode a superconductivity-specific T_c . It encodes a general *phase-transition energy scale*—some function of composition that tracks where a material’s dominant collective transition sits in temperature—and superconducting T_c is one readout of it. The model learned this without ever being told any material was magnetic: magnetic supervision was never part of any loss term.

Exploiting it: a trained Curie head. If the energy scale is really in there, a small head should read it out. The Curie MVP head described in the Methods section—pretrained with the encoder and all existing heads *frozen*, on the $\sim 19,000$ Curie-labeled examples of the expanded corpus—reaches **40.6 K eval MAE** (49.8 K train; 27 K in the <100 K bucket), against a published composition-based SOTA of 38.8 K test MAE from a dedicated gradient-boosted model (43). Two honesty constraints attach to this number. First, the experimental error floor for Curie temperatures is large: inter-laboratory spread on a single formula is routinely 30–80% (NdFe_2 is reported at 328, 453, and 578 K across independent measurements (43)), so a ~ 40 K MAE sits near the noise floor

of the labels themselves. Second, matching SOTA in aggregate is not a superiority claim—the NEMAD-derived labels were median/first-wins consolidated exactly as Jung et al.’s were, and no independent-source holdout has been run. What makes the result notable is not the 1.8 K gap to SOTA but the architecture: the same 2048-dim latent that supports superconductor generation, T_c regression, and family classification supports ferromagnetic Curie prediction with a two-layer readout—direct, quantitative evidence for the shared energy-scale hypothesis. The expansion roadmap this motivates (Néel, Debye, quantile-uncertainty heads; a universal phase-transition model) and the data engine it requires are discussed in Section 5.

4.7 Generative Holdout Recovery

The primary generative test is a Z-space search over the 429-target holdout (Section 4.8): for each target, the model generates hundreds of thousands of candidate formulas by exploring the latent space around element-anchored seed vectors—Z-vectors of training examples that share elements with the target—without being given the target’s composition, T_c , or Magpie features. It must discover a latent vector that decodes to the target purely from latent-space exploration. Full construction and per-candidate recovery for the holdout are given in the accompanying supplementary report; here I develop the search methodology and the model behaviors it exposes. The behavioral analyses in this section are computed on the final model (epoch 10,299, SHA d2027e24); the joint recovery counts themselves are reported in Section 4.8, where the last few targets are closed by a validity-head successor of near-identical latent geometry.

Z-space search methodology. The core strategy is element-anchored exploration: for each target, the model seeds from training Z-vectors sharing the target’s key elements and explores the surrounding neighborhood, augmented by (1) Gaussian perturbation at multiple scales, (2) spherical-linear interpolation (SLERP) between chemically similar Z-vector pairs, and (3) directed walks along principal components of the latent. Candidates are scored by a composite of element similarity and T_c proximity, where element similarity is the overlap of the target and generated molar-fraction vectors (1.0 = identical composition). The external recovery of Section 4.8 decodes exclusively with the two-stage decoder, which has no autoregressive drift by construction.

Zero hallucinated elements. Across the holdout the search never introduces an element absent from the target: the dominant—indeed essentially the only—error mode is stoichiometric-fraction imprecision, with the correct element set recovered even when trace-dopant fractions shift. This is the practically decisive property: the model reliably identifies *which elements to combine*, leaving only the proportions to refine.

Iso-formula T_c spread. Many Z-vectors decode to the *same* formula—single formulas are reached from thousands of distinct latent codes. A controlled perturbation study (1,500 Gaussian perturbations, $\sigma \in \{0.1, 0.3, 0.5\}$, around the best-recovery code of three exact-match targets) finds 210–431 perturbed codes decoding to the identical formula, yet the T_c head assigns them meaningfully different temperatures: a 3.8 K spread for $\text{Mg}_{9/10}\text{Li}_{1/10}\text{B}_2$, 11.5 K for $\text{NdFeAsO}_{17/20}$, and 18.3 K for a Bi-2223 variant. Since composition is fixed within each set this variation encodes something beyond composition, and it is physically ordered—smallest for the composition-determined MgB_2 -type material, largest for the multi-layer cuprate where structure matters most (62). A later training-data decomposition (Section 5, Figure 12) shows a substantial share of this spread is database/measurement scatter rather than a clean physical knob, so the non-compositional axis is best read as the model’s forward-prediction *uncertainty* at fixed composition—made precise by the causal analysis of Section 4.11.

Latent torsion. A local misalignment between the iso-formula and iso- T_c manifolds—moving along the direction that improves formula fidelity rotates away from the one that improves T_c —is partly by design: the decoder and T_c head are semi-independent readouts of one latent, and a `.detach()` on the T_c conditioning prevents them collapsing into a single computation. Its magnitude therefore measures how far training has aligned two objectives it never explicitly tied. Extended multi-task training contracted it sharply—the worst-case T_c disagreement anywhere in the holdout evaluation fell from 42.3 K at an earlier checkpoint to 13.5 K—without any explicit alignment loss. The residual is physically interpretable: the surviving high-error targets are Sb-substituted Bi-cuprates (whose T_c is apical-oxygen-sensitive (16)) and a rare delafossite, while composition-determined MgB_2 and 1111 pnictides (3; 2) sit at 0.0–0.3 K. The model in effect discovers *which materials need structural information* through the magnitude of its own internal misalignment.

From open search to a directed problem. The combinatorial space of plausible superconductor compositions (3–7 elements drawn from 118, each with fractional stoichiometry) exceeds 10^{10} candidates—the chemically-filtered quaternary space alone is $\sim 3 \times 10^{10}$ (42), and higher-order and fractionally-doped compositions grow it by orders of magnitude. For the recovered targets the search narrows this to the correct element set with approximately correct ratios, the correct family, and a T_c estimate within experimental noise—a mean absolute T_c error of 0.26 K (median 0.0 K; a handful of hard multi-dopant cuprates carry the tail, up to ~ 13 K), at or below typical measurement uncertainty for the high- T_c families where the model is most useful for discovery (64; 65). Discovery is thereby transformed from an open search over chemical space into a directed optimization within a specified compositional neighborhood.

4.7.1 Error Analysis

All 429 holdout compositions are ultimately recovered at similarity ≥ 0.90 (Section 4.8); what remains is a fraction-level gap between recovery and *exact* reproduction, and its structure is informative. The dominant residual error is stoichiometric-fraction imprecision, not element identity: of the recovered targets whose best candidate falls short of an exact match, the great majority differ from the target *only* in fractions (the element set is exactly right), a small number drop a single trace dopant, and *zero* introduce a hallucinated element. Three systematic patterns account for most fraction errors: nearby fractional representations competing in the output distribution (e.g. $\text{Sm}_{13/20} \rightarrow \text{Sm}_{67/100}$), renormalization to canonical unit-cell forms (arguably correct behavior—e.g. $\text{Y}_3\text{Ba}_5\text{Cu}_8\text{O}_{18}$ reduced to the Y-123 form), and equal-fraction site-sharing—the hardest pattern, since 50–50 occupancy is indistinguishable from single-element occupancy without crystal-structure information, now largely resolved ($\text{La}_{1/2}\text{Gd}_{1/2}\text{Ba}_2\text{Cu}_3\text{O}$ is recovered with both elements present rather than collapsing to one—though its T_c was the single residual target closed only by gradient steering in Section 4.8). Per-family recovery, exact-match, and T_c -outlier tables are in the supplementary report.

A note on family labels worth its own scrutiny: several targets the model labels CUPRATE_OTHER are dataset-labeled “Other” or “Conventional” yet contain Cu–O planes (e.g. $\text{Ba}_2\text{Ca}_5\text{Cu}_7\text{O}_{14}$); calling them generic cuprates is arguably *more* chemically correct than the training label, and treating such Cu–O-plane identifications as correct raises effective family accuracy substantially. The model has learned to identify cuprate chemistry even where its training labels do not reflect it—a first instance of the “generator as data auditor” pattern of Section 5.

4.8 External validation: joint composition-and- T_c recovery on 429 never-seen superconductors

Composition recovery (Section 4.7) is one axis; a stricter question—the one an experimentalist actually asks—is whether a recovered prediction is *usable*: does the model return both a synthesizable composition *and* a correct T_c ? To test this at scale we use the 429-target held-out set: superconductors drawn from databases disjoint from the curated training corpus, fingerprinted by rounded-fraction composition to isolate genuinely novel stoichiometries. None introduce an unseen element combination, so the test probes stoichiometric and T_c generalization *within* known chemistries. Its construction and provenance are detailed in the accompanying supplementary report.

We recover each target with the same element-anchored Z-space search (perturbation, SLERP, PCA-walk, plus orthogonalized element steering; Sections 4.7 and 4.9), decoding *exclusively* with the two-stage decoder, and score under a *joint* criterion: composition similarity ≥ 0.90 and $|T_c^{\text{pred}} - T_c^{\text{true}}| \leq \max(10 \text{ K}, 15\%)$. Because many latent codes decode to a single composition while predicting a range of T_c (the iso-formula variation of Section 4.7), the search is T_c -aware—among a target’s composition-matches it selects the code whose predicted T_c is closest to truth. This is legitimate for a recovery test, but it means composition alone does not fix T_c : the spread is the forward-prediction uncertainty for a genuinely novel material. Candidate budget is allocated adaptively: a light sweep of $\sim 4,000$ candidates per target (budget scale $B=1$) is run over all 429 and recovers most, while a heavy sweep of $\sim 48,000$ per target ($B=12$, a twelvefold budget) is spent only on the handful of residual failures. Under this protocol the base model (epoch 10,299) recovers 423/429 with both a synthesizable composition and a usable T_c ; the T_c requirement costs almost nothing (only 3/429 composition recoveries had an unreachable T_c). Its handful of residual targets are all complex multi-dopant cuprates (e.g. Hg–Sr–La–Ca–Ce–Cu–O and Gd–Pb–Sr–Ca–Cu–O systems)—the fine-

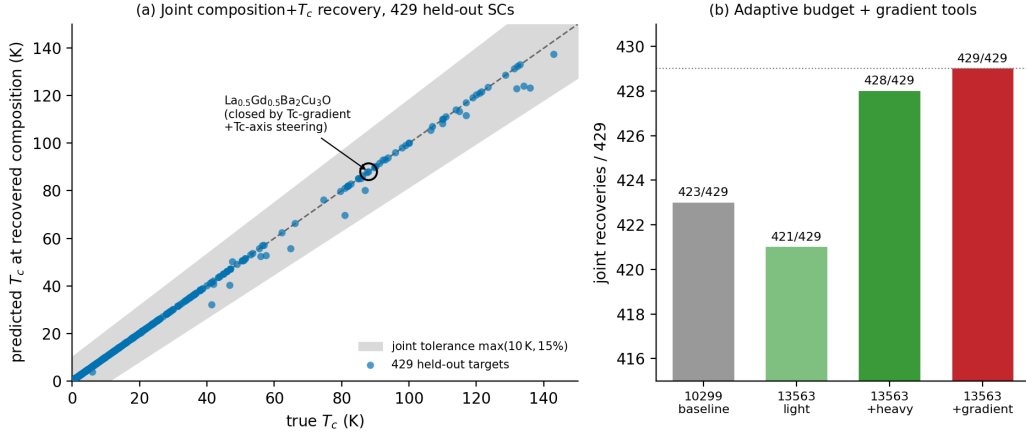


Figure 6: External joint recovery on 429 never-seen superconductors (validity-head successor). **(a)** Predicted T_c at the recovered composition vs. true T_c for all 429 held-out targets; the shaded band is the joint tolerance max(10 K, 15%). Every target lands inside it. The circled point, La_{0.5}Gd_{0.5}Ba₂Cu₃O, is the single target the sampling search left composition-correct but T_c -low; it is closed by T_c -gradient ascent plus T_c -axis steering. **(b)** Joint recoveries as budget and tools are added: the successor exceeds the base model’s 423/429 under the adaptive budget alone (428/429) and reaches 429/429 once the gradient tools are included.

stoichiometry regime that resists even the heavy budget—and, as the next paragraph shows, they are a search-coverage limit rather than a model limit.

A validity-head successor and T_c -aware gradient search recover all 429. These residual cuprates are a search-coverage limit, not a model limit. A successor checkpoint—the same architecture continued from this model’s lineage, with an added physical-validity head that scores whether a composition is chemically admissible—has a near-identical latent geometry: across the 30,936 compositions shared between the two models’ latent caches, the same material’s latent direction is preserved (mean cosine 0.985; 0.963 after removing a global rotation) and its predicted T_c drifts by a median of 0.005 K (100% within 5 K), so the interpretability findings of this paper transfer to it unchanged. Re-running the recovery on this successor gives 421/429 under the light budget and 428/429 after the same adaptive heavy sweep used above—already exceeding the base model’s 423/429; the sole remaining target—La_{0.5}Gd_{0.5}Ba₂Cu₃O, whose composition is recovered at similarity 0.992 but whose best composition-matching code predicts T_c 29 K low—is closed by augmenting the sampling search with the two gradient tools it had omitted: differentiable T_c -gradient ascent and stepping along the learned T_c -axis (Section 4.9). Together these locate a latent code that decodes to the correct La–Gd–Ba–Cu–O composition at its true 88 K (T_c error 0.0001 K), giving **429/429 (100%) recovery on both composition and joint T_c** (Figure 6). That the last target fell to differentiable steering rather than to more random sampling is the operative lesson: the material was always present in the latent space; the search simply lacked the gradient to reach it.

Recovery spans the full extent of the manifold. Recovery does not fall off with distance from the densely-sampled regions of composition space: the six holdout targets whose nearest training superconductor is compositionally most distant—in chemistries sparsely represented in training (lead chalcogenides, a Li–Pd–B boride, an yttrium carbide)—are *all* recovered, at composition similarity ≥ 0.98 and joint- T_c error ≤ 2.3 K. Coverage is therefore a property of the generative manifold across its full extent, not only its densely-sampled regions. The distance of each target from the training distribution, and the recovery-versus-distance breakdown, are given in full in the accompanying supplementary report.

4.9 Steering the Latent Space

Interpretability that cannot change model behavior is description; interpretability that can is control (32). This section closes the loop: the artifacts of Sections 4.4–4.5 become steering machinery for generation, and the steered generator produces a novel, database-verified candidate material.

Generation strategies. The generation pipeline explores the latent space with eight strategies: gradient ascent from non-SC seeds (S0, below); T_c -gradient boosting of known superconductors (S1); peak-targeting for cuprates using the extracted PySR quadratic expressions (S2); λ -directed search in the BCS region (S3); all-pairs cross-family interpolation, $\binom{N}{2}$ family pairs (S4); element substitution guided by a 72-element chemical-similarity map, inspired by GNoME (41) (S6); Sobol quasi-random space-filling exploration within per-family PCA subspaces (S7); and conditional T_c -targeted generation from random latent vectors (S8).

Strategy 0: turning cheap metals into superconductor candidates. The most conceptually direct use of the trained classifier is gradient ascent on $P(\text{SC})$ itself: start from the latent encoding of an inexpensive, manufacturable non-superconductor (Fe, Al, Ti, NbTi, FeSe, ...; 16 seeds), ascend $\nabla_{\mathbf{z}}P(\text{SC})$ until the classifier flips, then switch to $\nabla_{\mathbf{z}}T_c$ —while regularizing against the Magpie reconstruction head ($R^2 = 0.86$) so the walk stays on the manifold of physically coherent compositions and preserves the seed’s metallic/ductile character. This was impossible with a T_c -gradient-only pipeline: the T_c head is untrained in the non-SC region (Section 4.4’s contamination lesson), but the SC classifier’s gradient is trained *everywhere*. The pre-generation diagnostics validate the approach: the classifier’s decision boundary is nearly ideal ($P(\text{SC})$ gap of 0.9948 between non-SC and SC populations), and the decoder produces valid formulas for 86% of latent vectors even at perturbation scale $\sigma = 2.0$.

Interpretability-driven steering hooks. Four hooks consume the Section 4.5 artifacts directly: *ElementPusher* perturbs $\mathbf{z}$ along the $[2048 \times 118]$ element-correlation vectors to push candidates toward or away from specific elements; *DeadDimMasker* zeroes gradient components on dimensions the target head provably ignores; *HeadPCProjector* denoises gradients by projecting them onto the head’s top Jacobian principal components; and *BiasedClusterFilter* rejects candidates whose latent neighborhood had biased T_c predictions—in the final model this last hook is a no-op, because the error-landscape census found zero biased clusters.

Validity before novelty. Before a candidate reaches the (expensive) database search, it passes a deterministic basic-chemistry/physics gate—rule-based, *not* a learned head. The gate hard-rejects impossible chemistry: elements that cannot form bulk matter (superheavy $Z \geq 99$ and the transactinides, plus ultra-short-lived At/Fr), inert noble-gas constituents, absurd stoichiometry, and *family-inconsistent* decodings—a “cuprate” the decoder emitted without Cu or O. It soft-flags implausible-but-not-impossible cases (all-metallic “soups,” or a T_c above the realistic ambient ceiling for the predicted family—a conventional metal at 110 K). The gate is deliberately rule-based because the model’s *own* validity head is overconfident off the training manifold: it scored a nonsense tennesseine compound ($\text{Ts}_{0.5}\text{N}_{0.5}$ —an element with no bulk chemistry, and in fact a reproduced vocabulary-expansion artifact) at 0.97, and the six-database novelty check trivially labeled that same compound “novel” because no tennesseine compound is in any database. Only the deterministic gate catches it. This is the off-manifold overconfidence of Section 4.13 surfacing inside the generation loop—a *high head score is not evidence of a real material*—and it is why cheap rule-based validity runs first.

Novelty discipline. Generative-model “discoveries” are usually training-set reproductions wearing novel-looking formulas, and this pipeline initially produced exactly those: earlier candidates celebrated as novel (MgOFeSe_2 , LiBC, a Hg–Tl hybrid cuprate) were all found in the training data or its source databases once a deep novelty check was built. Novelty is therefore established in two stages before any claim is made. *First*, each candidate is checked against the training corpus itself by canonical composition—the strictest test, since a formula the model was trained on is a reproduction, not a discovery. *Second*, the survivors are cross-referenced against external references: six databases totaling $\sim 365,000$ entries (the training corpus, four NEMAD collections, and Materials Project), again matched on canonical composition, followed by a targeted literature search for the specific stoichiometry and its near-neighbors. Only a candidate absent from both the training set *and* every external reference is reported as novel.

A genuinely novel prediction: Hf-doped FeSe. Run at scale (two-stage decoding on the validity-head successor of Section 4.8), the gated, novelty-checked pipeline produces chemically plausible,

family-consistent candidates across every major superconductor family—doped iron pnictides, MgB_2 variants, cuprates, intermetallic alloys, and a hydride—catalogued with per-candidate checks in the companion candidate supplement; on a per-candidate literature search, that broader set resolves largely into rediscoveries of known superconducting chemistry, leaving the candidate below as the single fully-vetted survivor. The flagship is $\text{Fe}_{0.9}\text{Hf}_{0.09}\text{Se}$, predicted $T_c = 21.1$ K, from cross-family interpolation between the Heavy-Fermion and Iron-Chalcogenide latent regions (S4, $\alpha = 0.74$). It is absent from all six databases and from the literature—no published work on HfFeSe , $\text{Fe}_{1-x}\text{Hf}_x\text{Se}$, or Hf-doped iron chalcogenides was found. The prediction is a *hypothesis*, not a discovery: Hf’s metallic radius (~ 159 pm vs. Fe’s 126 pm) (79) makes Fe-site substitution challenging—and Fe-site substitution by transition metals, including the directly analogous Group-4 case Ti, is generally reported to *suppress* FeSe superconductivity (77; 78)—so 1–5% Hf is more synthesizable than the predicted 9%, and the enhancement over bulk FeSe should be read as an aggressive hypothesis a synthesis attempt could readily falsify; the nearest known material is FeSe itself ($T_c = 8$ K bulk (74), up to 65 K monolayer (75; 76)), so a 21 K prediction for a doped variant is aggressive but not absurd. A secondary candidate, $\text{BaFe}_2(\text{As},\text{Se})_2$ (Se-for-As substitution in the 122 pnictide, predicted 26.1 K), carries the same hypothesis-only framing. A subsequent 43,000-candidate coverage run (companion supplement) adds a further cross-family shortlist of gate-passing, database-novel hypotheses—the most credible a Co-doped Rb-122 pnictide, $\text{Rb}(\text{Fe},\text{Co})_2\text{As}_2$ -type ($T_c \approx 24$ K, the textbook doping route to 122 superconductivity (87)), alongside doped- MgB_2 , cuprate, and refractory-alloy variants—with one honesty caveat: these cleared the automated six-database novelty check but *not*, initially, the per-candidate literature search that Hf-doped FeSe received. A subsequent documented literature search of the shortlist (companion supplement) found *no* additional clean survivor—the candidates resolve into rediscoveries of known superconductors, known non-superconductors, a magnetic false positive, and chemically-implausible artifacts—so the flagship remains the single deeply-vetted prediction. The full shortlist, with each candidate’s novelty check, head predictions, and an economic-viability assessment, is in the companion supplement; careful experimental synthesis-route design is left to future work.

Calibration caveats on generated T_c values. Several other generated candidates carry T_c predictions that are almost certainly inflated (e.g., a 122-type pnictide variant at 198 K where the family caps at ~ 38 K (83), and a transition-metal sulfide at 166 K whose prediction likely reflects Cr-Fe-S *magnetic* transitions at 160–170 K (84) rather than superconductivity—the phase-transition energy scale of Section 4.6 leaking into the T_c readout). These are flagged and excluded from any claim—the 198 K pnictide is exactly the case the validity gate’s family-appropriate- T_c ceiling now catches automatically, and the magnetic-transition confusion is the same failure mode the phase-transition heads are designed to disambiguate.

The generator as a data auditor. Chasing a suspicious 200 K candidate led back to its training neighborhood and exposed a mislabeled training example: $\text{Ca}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$, a spinel ferrite recorded as a 200 K superconductor in a source database—its “transition” is magnetic ordering, not superconductivity. The label was corrected. Together with the mislabeled-Curie materials of Section 4.6 and the Cu-O-plane classifications of Section 4.7.1, this is a recurring pattern worth stating plainly: *a sufficiently calibrated model disagrees with its training data exactly where the data is wrong*, and those disagreements are auditable.

4.10 Causal Validation: From Correlation to Control

The *ElementPusher* hook of Section 4.9 perturbs $\mathbf{z}$ along the *correlational* element vectors of Section 4.5. Whether those directions are *causal*—and whether the atlas’s concentrated per-dimension encodings (oxygen on dimension 276, copper’s bipolar 888/436) actually localize element identity—is a separate question, and the interpretability literature warns that per-feature attributions can overstate localization (40). We answer it with direct latent interventions—single-dimension ablation and steering along the correlation vectors, reading off the effect on generation—in the spirit of interchange interventions and activation steering (66; 33); the fuller causal toolkit of path patching and causal scrubbing (67; 68) we approximate with steering rather than run in full. I test the atlas with interventions: for each element E I ablate (mean-substitute) or steer its atlas dimension(s) and measure the effect on the model’s own outputs—the encoder’s element-presence head and the autoregressive (AR) decoder, decoded greedily so any change is purely the intervention.

Hydra MAGE: element identity is a distributed linear direction — single-dim ablation is null, full-vector steering is strong

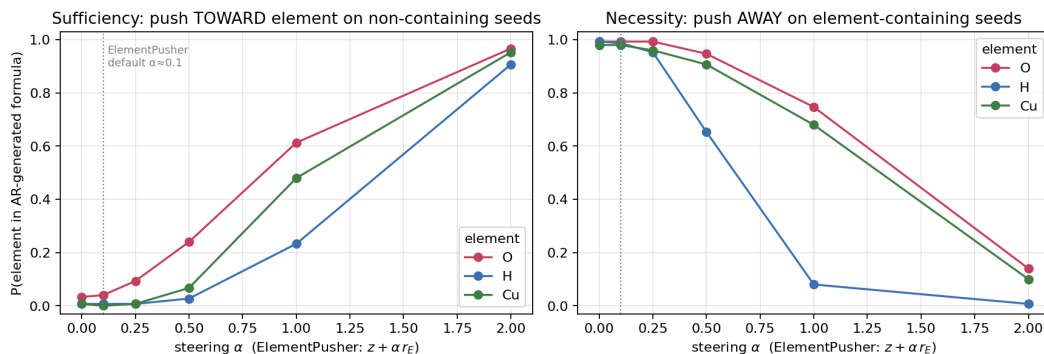


Figure 7: Causal steering of AR generation along the full element-correlation vector $\mathbf{z} + \alpha \mathbf{r}_E$. Left: sufficiency (pushing toward an absent element); right: necessity (pushing away from a present one). Single-dimension ablation (not shown) is null; the coordinated direction is strongly, monotonically causal. The dotted line marks the prior default step $\alpha \approx 0.1$, far too weak; the effective range is $\alpha \approx 0.5$ –2.

Single dimensions are not causal; the direction is. Ablating an element’s top-correlated dimension shifts the readout head for hydrogen ($\Delta \logit = -1.81$) and oxygen (-0.66), but not copper—whose bipolar encoding (dimensions 888/436) is individually inert—nor iron. On the AR decoder, single-dimension ablation is null for *every* element: sweeping oxygen’s dimension 276 across $\pm 3\sigma$ leaves the oxygen content of generated formulas flat at 6.1–6.2 atoms. Yet steering along the *full* correlation vector, $\mathbf{z} + \alpha \mathbf{r}_E$, moves generation strongly and monotonically (Fig. 7): oxygen presence rises from 3% to 97% and copper from 1% to 95%. Element identity is a *distributed linear direction*, not a localized dimension—invisible to single-neuron ablation but linearly steerable. The same holds for T_c : ablating the top-gradient T_c dimensions causes $5\times$ more change than low-gradient dimensions (gradient attribution is causally meaningful), yet every single-dimension effect is negligible— T_c is computed over a redundant subspace. This is a clean instance, outside the language-model setting, of the superposition the mechanistic-interpretability literature documents (37): the concentrated correlations of Section 4.5 are real but do not imply single-dimension causality—a materials instance of the broader caution that a component’s ablation-necessity or linear-decodability does not by itself establish a mechanistic role (36).

Steering directions encode learned chemistry. Measuring how every element’s generated abundance changes when steering toward a single element yields a specificity matrix (Fig. 8) whose off-diagonal structure recovers real superconductor families: steering toward Cu drags in O ($\Delta +1.1$ atoms) and Ba (cuprates); Fe drags in As (iron pnictides); oxygen is a hub, suppressed by every non-oxide push. The element directions are entangled by chemical co-occurrence, not pure identity.

Orthogonalized steering enables off-pattern generation. Projecting the co-occurring direction out of the target (Gram–Schmidt, renormalized) de-entangles the code: orthogonalized copper steering raises Cu while *removing* O ($+2.2$ vs. -1.4 atoms), producing valid oxygen-free copper intermetallics that raw steering cannot reach; orthogonalized iron steering yields iron chalcogenides rather than pnictides. This is a concrete generation-control primitive, and it explains why naively steered “novel” candidates tend to collapse back onto the dominant cuprate manifold: to reach genuinely off-distribution chemistry, one must steer along an element direction with its co-occurring partners projected out.

4.11 What the T_c readout computes: a stored coordinate, not a mechanism

Causal intervention sharpens the template-fit caveat of Section 4.4: when the model returns a T_c , is it *computing* it from a mechanism, or reading it off a coordinate? Three interventions give the latter.

T_c is causally decoupled from the model’s own phonon scale. A linear probe recovers a Debye temperature θ_D from $\mathbf{z}$ at $R^2 = 0.77$ (metallic subset; permutation-null ≈ 0), yet the T_c head does not route through it. Steering $\mathbf{z}$ along the θ_D direction—composition preserved (97% by

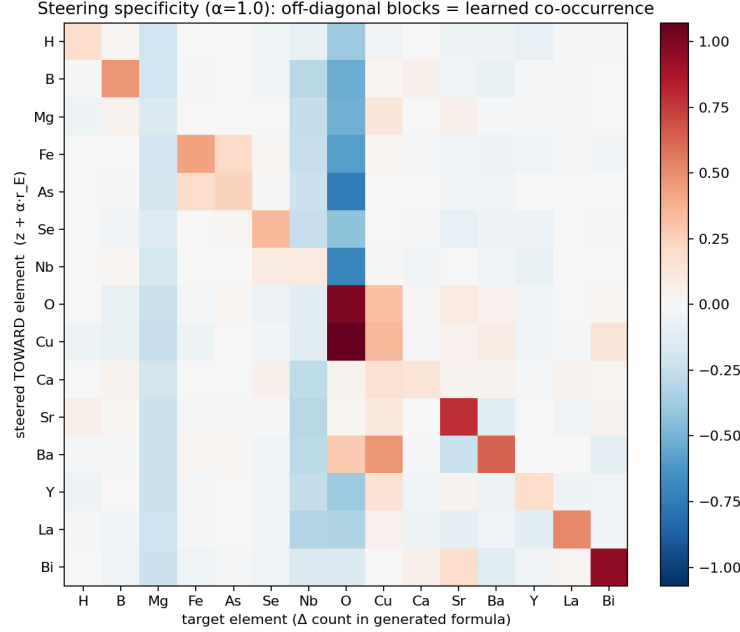


Figure 8: Steering specificity: change in each target element’s generated abundance when steering toward each source element. Off-diagonal blocks recover the model’s learned chemistry—the cuprate O–Cu–Ba block and the iron Fe–As block—showing element directions are entangled by co-occurrence.

decoded-formula check)—moves the model’s own θ_D five-fold while its T_c prediction does not move: the causal prefactor $\partial \ln T_c / \partial \ln \theta_D = 0.00$ (bootstrap CI $[0, 0]$), against the $+1$ that a BCS form $T_c \approx 1.14 \theta_D e^{-1/N(0)V}$ (7) requires. A positive control (steering the orthogonal T_c -driver direction *does* move T_c) and a coupling control (the null survives holding a probed electron–phonon λ fixed; θ_D and λ are near-orthogonal internal directions) rule out artifacts. The model has the BCS ingredients but computes T_c around them.

T_c is a reconstructed input. The encoder is given T_c ; sweeping that input at fixed composition, the head’s output tracks it almost exactly ($\partial T_c^{\text{pred}} / \partial T_c^{\text{in}} = 0.96$, deterministic), and ablating the input to a neutral value collapses composition-only reconstruction below the mean ($R^2 < 0$). Composition is only weakly T_c -predictive in principle here—a *linear* probe of the Magpie composition features reaches only $R^2 = 0.38$ on the hard conventional-superconductor subset (conventional T_c needs structure the model never sees), well below the nonlinear $R^2 = 0.92$ full-corpus Magpie baseline reported earlier, which is inflated by the mass of trivially zero- T_c non-superconductors—so the head reconstructs the stored value rather than inferring it. This is *why* no composition descriptor aligns with the T_c head’s causal driver, and why its T_c function is not BCS-shaped.

The reverse lane is a live but weakly-physical doping knob. The two-way street is real generatively: conditioning the decoders on a swept T_c (fixed $\mathbf{z}$) deterministically changes generated *stoichiometry* at fixed elements, a directional response carried by the two-stage fraction decoder ($\sim 10\times$ more T_c -responsive than the composition-locked autoregressive decoder, which also explains the cross-decoder disagreement used as an out-of-distribution signal in Section 4.13). This is the physically right variable—doping tunes cuprate T_c at fixed elements—but only weakly the right direction: it fails to move the textbook dopant knobs (e.g. Ca in multi-layer cuprates, where the training data itself carries a $+0.6$ T_c –Ca correlation), leaving a faint aggregate physical footprint ($+0.41$). The conditioning lane is active but not grounded—the target of the doping-regularization program of Section 5.

Together these make the effective-theory reading precise on *causal* rather than correlational grounds: Hydra MAGE holds a joint (composition, T_c) representation it interpolates smoothly—a navigable coordinate, not a physical T_c mechanism.

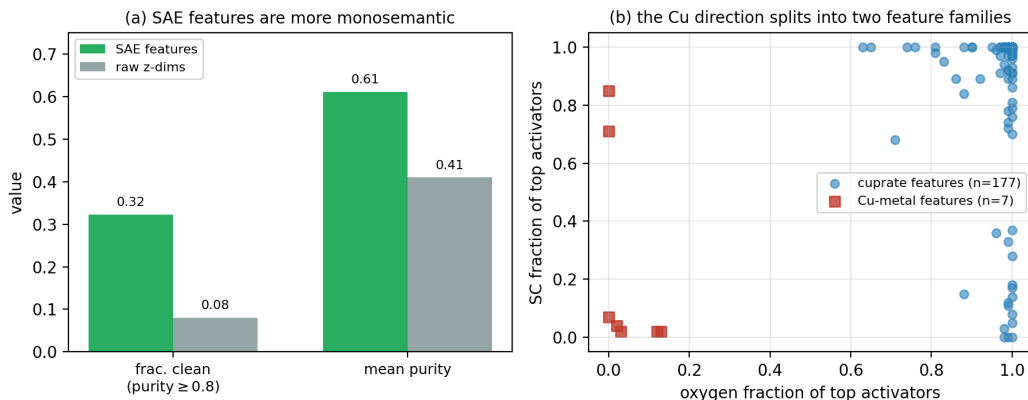


Figure 9: Sparse-autoencoder decomposition of the latent. **(a)** SAE features are more monosemantic than raw latent dimensions—a $4\times$ higher fraction reach purity ≥ 0.8 (32% vs. 8%), at higher mean purity (0.61 vs. 0.41). **(b)** Among the 185 Cu-rich features, oxygen content splits them into two near-independent families (activation correlation 0.03): oxygen-rich cuprate features whose top activators are predominantly superconducting (upper right) and 7 oxygen-poor copper-metal features whose activators are non-superconducting (lower left)—the entangled “Cu $\Rightarrow$ cuprate” direction of Section 4.10, decomposed automatically.

4.12 Sparse Decomposition of the Latent Space

If element identity is a distributed, entangled direction, is there a basis in which the latent’s features become monosemantic? I train a Top- K sparse autoencoder (SAE)—the dictionary-learning approach that has become central to language-model interpretability (29; 30) and is beginning to reach other modalities (38)—on the 60,570 cached latent vectors ($2048 \rightarrow 4096$ dictionary, $k = 32$ active features, reconstruction $R^2 = 0.95$). SAE features are substantially more monosemantic than the raw latent dimensions: 32% reach purity ≥ 0.8 (a single dominant element among their top activators) against 8% of raw dimensions, and mean purity rises from 0.41 to 0.61 (Fig. 9a). Clean element- and-property features emerge—a mercury–oxide superconductor feature, an ytterbium–oxide feature, pure single-element features. Most tellingly, the entangled “Cu $\Rightarrow$ cuprate $\Rightarrow$ high- T_c ” direction of Section 4.10 *splits* into two near-independent feature families (Fig. 9b): of the 185 Cu-rich features, 177 are oxygen-rich cuprate features (Cu+O, top activators predominantly superconducting, high T_c) and 7 are oxygen-poor copper-metal features (Cu, no oxygen, top activators non-superconducting, $T_c \approx 0$), with activation correlation 0.03 and top-activator overlap 0.02. The model represents copper-in-a-cuprate separately from copper-in-a-metal; the distinction is superposed in the raw latent so no single dimension exposes it, and the sparse basis recovers it automatically—the unsupervised analogue of the orthogonalized steering above.

4.13 Confident Failures Off the Manifold

Pushing generation off the training manifold exposes a calibration failure (69; 70)—the materials analogue of a classifier’s confident predictions on unrecognizable inputs—quantified here on a 43,000-candidate coverage atlas (companion candidate supplement) binned by latent distance to the nearest training vector (Fig. 10). As the figure shows, the model’s explicit confidence heads (competence and $P(\text{SC})$) do not track the distance, staying high across the off-manifold range where candidates actually land, while two *other* signals do: the emergent disagreement between the AR and two-stage decoders reading the same latent, and the possibility-head of the validity-head successor (Section 4.8). That added head is thus precisely the one that carries the out-of-distribution signal the original SC and competence heads ignore—though it is itself leaky (43% of chemically-impossible generations still score above 0.7), which is why the deployed pipeline (Section 4.9) filters with a deterministic chemistry gate rather than the learned score. Auditing the confident off-manifold positives sorts them into two mechanistically distinct buckets. *Structure-blind* failures are chemically real but structurally under-determined: a predicted 80 K FeS is a genuine iron chalcogenide whose bulk T_c is only ~ 5 K (81), yet the closely-related monolayer FeSe reaches 65–100 K (82)—composition

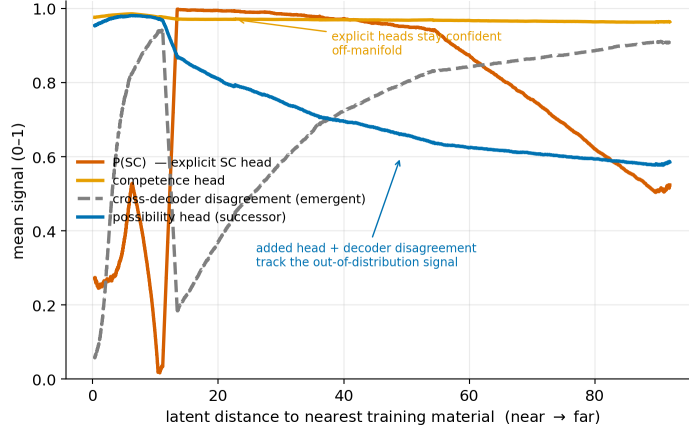


Figure 10: Off-manifold overconfidence, on a 43,000-candidate coverage atlas (companion candidate supplement) binned by latent distance to the training manifold (near $\rightarrow$ far). The explicit confidence heads do not track distance: the *competence* head stays flat-high (~ 0.97) everywhere, and $P(\text{SC})$ is calibrated only immediately adjacent to training—where it correctly tracks the $\sim 57\%$ non-superconducting neighbourhood—then saturates high across the off-manifold range. Two *other* signals do track it: the emergent cross-decoder (AR vs. two-stage) disagreement climbs, and the *possibility*-head of the validity-head successor (Section 4.8) declines monotonically from 0.96 to 0.58. The out-of-distribution signal the explicit heads ignore is carried instead by the decoder-ensemble disagreement and by the single head added to the successor.

alone cannot fix which structural regime applies, the same hidden-variable limit that produces the residual latent torsion of Section 4.7. *Family-shortcut* failures are the diagnosable ones: the model fires “element $\Rightarrow$ family $\Rightarrow$ high- T_c ” on a marker element even when the family’s defining partner is absent—a 144 K Cu–Pd–Zn alloy (copper without the cuprate oxygen, and palladium is a paramagnon-suppressed non-superconductor (85)), a 79 K “MgB₂-type” Al–Mg–Y with no boron—or ignores magnetic pair-breakers. This is the entangled element direction of Section 4.10 seen from the failure side: activating the marker without its partner still triggers the whole family prediction.

The interpretability findings suggest a targeted fix rather than merely a caveat. Keeping the black-box T_c head, one can add a second, physics-grounded estimate T_c^{phys} computed through explicit closed-form theory (Allen–Dynes for the phonon channel, family bounds gated on the required constituents)—so that a composition lacking cuprate oxygen or MgB₂ boron cannot inherit that family’s T_c by construction. Their disagreement $\Delta = |T_c^{\text{DL}} - T_c^{\text{phys}}|$ is then a calibrated grounding signal: a prototype gives median $\Delta \approx 8$ K on real superconductors versus ≈ 90 –110 K on the confident false positives (Cu–Pd–Zn: $T_c^{\text{phys}} \approx 0$ against the head’s 144 K). Interpretability that localizes a failure mechanism, rather than only measuring its rate, turns directly into an intervention—the same discipline as the “generator as data auditor” of Section 4.9, applied to the model’s own predictions.

Tracing and editing the shortcut circuit. The family-shortcut is not merely diagnosable but *localizable and correctable* at the feature level, following the sparse-feature-circuit and SHIFT methodology (31). Because the confident false positives occupy a latent region one can enter directly, the shortcut is realizable as a controlled counterfactual: zeroing the oxygen-partner SAE features (Section 4.12) in a real cuprate’s code—stripping the oxygen evidence while leaving the copper-marker features intact—produces a copper-without-oxygen positive on which the SC head’s confidence *survives* ($P(\text{SC})$ falling only to 0.57–0.88 even as predicted T_c collapses), the shortcut caught in the act. Per-feature ablation attribution then isolates the ~ 14 –20 residual features that still drive the SC readout (copper-, barium-, bismuth-, and palladium-dominated marker and family features); ablating this attributed circuit collapses the false positive ($P(\text{SC}) \rightarrow 0.04$), while the *same* ablation applied to intact cuprates leaves them superconducting ($P(\text{SC}) = 0.99$)—a surgical, SHIFT-style removal of the spurious pathway that spares the legitimate one. The result replicates on

the validity-head successor ($0.27 \rightarrow 0.002$, specificity 0.93). Two honesties qualify it. The circuit is *distributed*: the semantically tagged copper-marker features alone do not fix the false positive; only the attribution-identified set does—the superposition caution of (36; 37) made concrete. And it is a circuit in the *readout* representation, not a claim of input-level necessity: a composition-space counterfactual that instead removes oxygen from the input and re-encodes leaves the SC head saturated (logit ≈ 41 , $P(\text{SC}) = 1.0$), because the encoder re-maps the de-oxygenated cuprate back onto the superconducting manifold from its cation skeleton. The ablation localizes the oxygen-dependent *readout direction* the head uses—a different, weaker statement than the model requiring oxygen in the formula, but enough to suppress the confident false positive at inference: the circuit-level analogue of the Δ grounding signal above.

4.14 From Latent Discovery to Deployable Candidates

A candidate superconductor with spectacular T_c but unmanufacturable chemistry is a paper, not a product. To rank generated candidates by deployability, each receives a composite economic score (0–100) built from five components: raw material cost (USGS 2025 mineral commodity prices (86)), cooling tier (helium-crisis-aware: liquid-nitrogen and cryocooler regimes score far above liquid-helium), processing complexity (simple alloys score highest, complex oxides lowest), supply-chain risk (weakest-link element risk from production volume, geopolitical concentration, and byproduct status), and a high-pressure penalty driven by the HP and pressure heads.

Benchmarking the score on known superconductors reproduces industrial reality: MgB_2 scores 74.5 (cheap, cryocooler-compatible) and NbTi 68.5 (the workhorse MRI conductor), while YBCO scores 54.8 and Bi-2223 54.0—high- T_c but expensive and hard to process—and Tl-2223 falls to 42.6 on toxicity-adjacent supply risk. That the score independently ranks the materials industry actually deploys above the materials it mostly does not is the sanity check that makes it useful as a generation filter.

5 Discussion and Future Work

What the model learned. The analysis reveals a latent representation that encodes family identity, T_c , and the SC/non-SC phase boundary in a geometrically organized manifold. Family-specific gradient profiles show the model applies different computational pathways to BCS metals, cuprates, iron-based materials, and MgB_2 —functioning as a multi-theory effective representation rather than a single compositional shortcut. The model-internals program of Sections 4.4–4.5 makes that claim concrete: the sub-theories can be written down as per-family equations, their effective dimensionalities measured, and their element encodings traced to nameable latent dimensions—though the causal analysis of Section 4.10 shows those concentrated correlations are not individually causal: element identity is a *distributed* linear direction, steerable only as a whole. The holdout evaluation demonstrates that the representation generalizes: across the 429-target holdout the correct element set is recovered with zero hallucinated elements and residual errors confined almost entirely to stoichiometric fraction imprecision. The semi-independence of the decoder and T_c heads is by design: they share a backbone and latent space but optimize different objectives. The residual latent torsion between their manifolds contracts on its own over continued training (quantified in Section 4.7), without any explicit alignment loss—evidence that the architecture converges toward aligned solutions, with self-consistency losses available to accelerate what training is already doing.

Generative evaluation via Z-space search. The Z-space search is the central generative evaluation: the model must produce holdout formulas without receiving any target information, purely by exploring the latent space around element-anchored seed vectors from training data. It recovers all 429 held-out superconductors on both composition and T_c (Section 4.8), and dramatically outperforms a roundtrip encode–decode baseline—demonstrating that the learned latent space contains never-seen superconductors at navigable locations the encoder itself cannot reach.

Implications for superconductor discovery. For high-confidence families (MgB_2 , Hg-cuprate, YBCO, Iron-based), generated candidates can be trusted to have correct elements and approximately correct stoichiometry, with family-level T_c MAE of 0.0–0.8 K. For medium-confidence families (Tl-cuprate, Conventional), candidates have correct elements but may need experimental verification of exact stoichiometry. The residual risk regions are Sb-substituted bismuth cuprates and structurally rare compounds (delafossites), where T_c calibration lags formula fidelity.

Could 1.24 million candidates find holdout matches by chance? The search evaluated $\sim 261,000$ Z-vectors per target, decoding to a mean of $\sim 27,500$ unique formulas per target. A naive estimate of the chemical composition space is instructive: with 118 elements and typical superconductor formulas containing 3–7 elements with fractional stoichiometries, the combinatorial space of chemically plausible compositions is astronomically large ($> 10^{10}$ even restricting to known superconductor-like patterns). Recovering the full holdout from $\sim 27K$ unique candidates per target by random sampling would require extraordinary luck. More concretely, the search is not random: it is seeded from element-anchored Z-vectors of chemically similar training materials and explores the *local* latent neighborhood around those seeds. The high recovery rates therefore reflect the latent space’s organization—chemically similar materials map to nearby Z-vectors, and the decoder produces chemically coherent outputs in those regions. The fact that zero extra elements were hallucinated (the model never adds elements that don’t belong) further argues against chance: a random walk through Z-space would produce arbitrary element combinations, not consistently correct element sets with only minor stoichiometric errors.

Evaluating a generator, not a regressor. The holdout evaluation deserves explicit methodological justification. In standard T_c prediction work, models are evaluated by a train/validation/test split (typically 80/10/10), with test-set MAE or R^2 as the metric. A 20% test split from $\sim 26,000$ SC samples would yield $\sim 5,200$ test materials—statistically powerful but testing only whether the model can *predict a number* from a composition. The present evaluation asks a fundamentally harder question: can the model *generate* a never-seen composition from latent space exploration alone, with no target information as input? This is not a regression test but a combinatorial search problem. Each held-out target is a specific needle in a haystack of $> 10^{10}$ chemically plausible compositions, and the model must find it by exploring $\sim 27,000$ unique candidates in a 2048-dimensional space. Recovering them under these conditions is substantially more demanding than predicting T_c for thousands of known compositions, because the model must simultaneously get the element set, all stoichiometric fractions, the T_c , and the SC classification correct—all without seeing any target data.

The difficulty is compounded by a fundamental aspect of superconductivity: the phenomenon arises from multiple mutually exclusive physical mechanisms. BCS/Allen-Dynes phonon-mediated pairing governs elemental and A15 superconductors (7; 53); MgB_2 requires a two-gap strong-coupling framework (3); cuprate superconductivity involves *d*-wave pairing mediated by spin fluctuations (8); iron pnictides exhibit $s_{\pm}$ -wave pairing from inter-band nesting (2); and heavy-fermion and organic superconductors involve still other mechanisms. The model must implicitly learn *which mechanism applies* from composition alone—without explicit labels for the pairing symmetry, mediating boson, or order parameter—and correctly predict T_c under the appropriate physical framework. The holdout spans all of these mechanisms: conventional BCS metals, MgB_2 variants, four cuprate sub-families, iron pnictides, and miscellaneous compounds. That the model recovers targets across this mechanistic diversity, with family-specific T_c accuracy (detailed in the supplementary report), provides evidence that the latent space has organized itself into a multi-theory representation. The holdout is drawn from databases disjoint from the training corpus (Section 4.8), giving a large test set without sacrificing training coverage; further curated sets (SODNet) remain to be added.

Limitations. The $R^2 = 0.9999$ is on training data; the generative evaluation covers 429 held-out targets within known chemistries, which measures recovery/coverage across families and mechanisms rather than extrapolation to novel chemistry. The composition-only input means structural variables can only be inferred statistically. The dominant error mode is stoichiometric fraction imprecision (29 of 31 non-exact recoveries), and residual latent torsion leaves four targets with > 10 K T_c error despite high compositional similarity. The extracted equations are lossy summaries of the network’s function (R^2 down to 0.65 for YBCO) and carry the switching-direction caveat of Section 4.4. Generated-candidate T_c values can be inflated outside well-populated regions (Section 4.9), and the Curie head’s ~ 40 K MAE sits near the label-noise floor rather than beneath it. No physical constraints (charge neutrality, thermodynamic stability) are enforced during generation in the base model; the validity-head successor of Section 4.8 adds a chemical-admissibility head as a first step toward this.

Future data integration. Several additional databases have been staged but not yet integrated: SuperCon2 ($\sim 40K$ NLP-extracted records from 37,700 papers), 3DSC (1,578 superconductors with *resolved crystal structures*, a subset of the 16,414-entry composition list (15)), SODNet (curated with verified T_c), and further DFT-predicted JARVIS structures beyond the hydrides already ingested. Rather than training on all new data uniformly, the model’s current competence suggests a curriculum strategy: generate candidates from the new databases, identify which ones the model

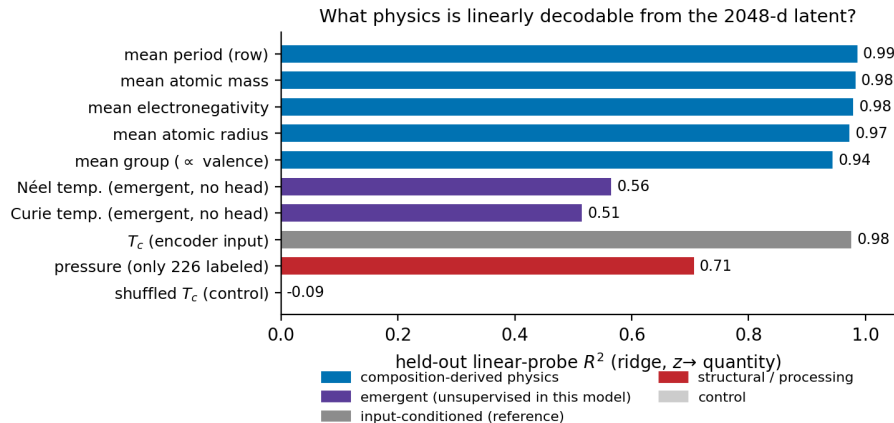


Figure 11: Linear-probe sweep: held-out R^2 of a ridge regression decoding each physical quantity from the 2048-d latent z (9,000 sampled materials; epoch-10299 model). Composition-derived physics—electronegativity, atomic mass and radius, valence/group, period—is linearly encoded at $R^2 = 0.94$ – 0.99 , far above a label-permutation control (shown for T_c ; matched-sample-size permutation nulls for every probe, including the $N=226$ pressure and $N \approx 2900$ magnetism probes, are all $R^2 \leq 0$, ruling out high-dimensional overfitting), confirming the latent is a faithful, linearly-readable composition representation. More tellingly, Curie and Néel transition temperatures—for which *this* model has *no head* and which it never saw in training—are recoverable at $R^2 = 0.51/0.56$ (2,843/2,931 composition-matched materials): magnetism is *emergently*, if partially, present in a latent shaped only by superconductivity and composition. The encoder-input T_c is recovered at 0.98 (a reference, since it is fed in); the one structural variable with labels (pressure, 226 entries) is only weakly recoverable (0.71)—the boundary of what composition determines.

struggles to reproduce or predict accurately, and selectively train on those failure cases. This targeted augmentation—already demonstrated in the iterative addition of rare-element non-SC materials during initial training—would maximize the marginal value of each new training sample. Incorporating 3DSC structural data could address the hidden variable problem, but must be done carefully: structural data availability is inhomogeneous across families, and selectively including it risks introducing training bias.

Deepening the interpretability program. In the March version of this paper, extracting the learned effective theory as explicit equations was the headline future-work item; Sections 4.4–4.5 now report it done, audited, and in one case retracted-and-corrected. The frontier has moved to *semantics*: linear probes tying gradient-PC scores and fitted latent readouts (the Presland $p(z)$ of Section 4.4, the Allen-Dynes $\lambda(z)$) to measurable physical variables—doping level, coupling constants, Debye temperatures. A direct $z \rightarrow$ quantity linear-probe sweep (71; 72)—validated with permutation-null control tasks in the sense of Hewitt and Liang (35) (Figure 11)—confirms that composition-derived physics is linearly encoded at $R^2 = 0.94$ – 0.99 , far above the control, and—more tellingly—that Curie and Néel temperatures probe at $R^2 \approx 0.5$ from a latent with *no* magnetic head, a quantitative linear-probe version of the emergent phase-transition scale of Section 4.6. But composition-*derivable* quantities being decodable is necessary, not sufficient, for rediscovered physics; it is exactly the sub-type-interpolation shortcut the Vafa et al. (40) caveat warns about. The discriminating probes are the *non-compositional* ones—coupling constants λ , doping level, crystal structure—which remain label-starved; supplying those labels (the data engine below) is what would separate a genuine effective theory from a merely faithful composition encoder. A complementary direction moves from representation-level interpretability to explicit *feature circuits*: the sparse-feature-circuit and SHIFT methodology (31) of tracing and editing causal graphs of SAE features, which Section 4.13 now applies to the marker-element $\rightarrow$ family $\rightarrow T_c$ shortcut—attributing and ablating the residual feature circuit to suppress its confident false positives while sparing real cuprates. Scaling this from a single audited shortcut to a full causal graph across the failure taxonomy, and from readout-level ablation to input-grounded intervention, is the natural next step.

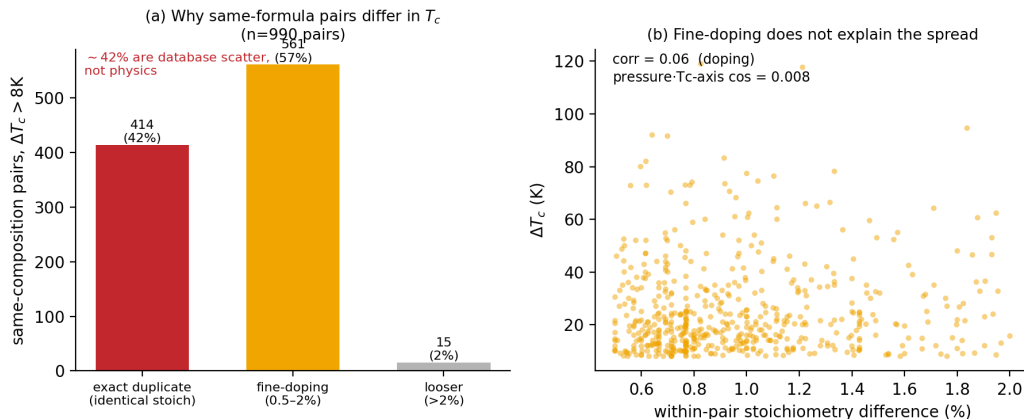


Figure 12: Is the “same formula, different T_c ” spread physical? **(a)** Decomposition of 990 same-composition training pairs whose T_c differs by >8 K: $\sim 42\%$ are *exact duplicates* (identical stoichiometry, different reported T_c)—database/measurement scatter, not physics. **(b)** For the remaining fine-doping pairs, the within-pair stoichiometry difference does not predict the T_c difference (correlation 0.06), and the T_c -axis is orthogonal to a labeled pressure direction ($\cos 0.008$). The residual, non-compositional T_c variation is therefore dominated by scatter and unlabeled structure rather than a controllable physical parameter—motivating a distributional, physics-grounded T_c head.

Reining in the T_c head: from a conditioned point estimate to a grounded distribution. The generative results above lean on the T_c head, and it is worth stating plainly what that head does and does not yet do well. *Strengths*: conditioned on the encoder’s inputs it is highly accurate ($R^2 = 0.9999$), it drives the joint recovery of all 429 external targets, and its per-family accuracy (detailed in the supplementary report) shows it applies mechanism-appropriate readouts rather than a single compositional shortcut. *Weaknesses, now measured*: first, the iso-formula T_c spread that the recovery exploits (Section 4.8) is largely *not* a physical hidden-variable knob. Decomposing 990 same-composition/different- T_c training pairs, $\sim 42\%$ are *exact-duplicate* database entries—identical stoichiometry, different reported T_c —i.e. source/measurement scatter rather than physics; the fine-stoichiometry difference within a pair does not track the T_c difference (correlation 0.06); and the learned T_c -axis is nearly orthogonal to a linear pressure direction fit on the 226 pressure-labeled materials (cosine 0.008, though pressure is too data-starved here to be conclusive; Figure 12). The bulk of T_c that the model gets right is *compositional* and genuinely physical; the residual, non-compositional axis is dominated by database noise and unlabeled structure, so the spread should be treated as forward-prediction *uncertainty*, not as a controllable physical parameter. Second, the head is a point estimate not tied to any theory, and at the frontier it can miss (the hardest holdout cuprate was under-predicted by 29 K before gradient steering reached it). For *discovery* this is not disqualifying: the task is asymmetric—a single genuine candidate is a success, and a mis-stated T_c on any individual candidate is recoverable in post-processing by an independent check—so the value of the head lies less in per-sample accuracy than in what its behavior reveals about the physics the model has internalized. The deeper objective is therefore to understand the *effective theory* the model—the T_c head included—is actually using, and to make that reasoning auditable rather than merely accurate. *The plan to rein it in* has three parts: (i) make T_c *distributional* via quantile/pinball regression (reusing the calibrated-interval machinery already built for the Curie and Néel heads), so a composition maps to a T_c *range* whose upper quantile is the phase-boundary—the discovery-relevant maximum—and whose width honestly absorbs the database scatter; (ii) *ground it in theory* with a mechanism-gated dual estimate—tie the learned T_c to a closed-form physics T_c (Allen–Dynes/Presland) in the phonon-mediated regime and penalize disagreement, using $\Delta = |T_c^{\text{DL}} - T_c^{\text{phys}}|$ as a grounding-and-anomaly signal while leaving unconventional families free but flagged; and (iii) since the context axis cannot be grounded from the latent alone, supply *external* structure and pressure labels (the data engine below) to give it physical meaning. The goal is a T_c readout whose values carry calibrated, mechanism-aware confidence—so that a high generated T_c is either theory-consistent or explicitly flagged as off-theory, rather than a bare number.

From superconductivity to a universal phase-transition model. The emergent energy scale of Section 4.6 sets the research direction I consider most consequential, and it can be stated as a single, strange, but precise question: *how much physically valid information can be extracted from this 2048-dimensional composition representation?* Each frozen-encoder readout head is an experiment on that question. The latent was shaped only by superconductivity tasks and compositional reconstruction, yet a two-layer readout recovers Curie temperatures near the label-noise floor. So the program is: keep training small readout heads for additional physical properties—Néel temperature, Debye temperature, melting point, and beyond—*until they fail*. Every head that succeeds proves the corresponding physics was already implicit in composition-plus-multi-task structure; the first properties whose readouts fail map the boundary of the representation, separating what composition determines from what genuinely requires structural, electronic, or processing information the encoder never saw. Either outcome is informative: success extends the model, and failure localizes, property by property, the information that a formula alone cannot carry.

The designed vehicle for this program—the *universal phase-transition model*—keeps the shared composition encoder and adds gated classifier-regressor pairs per transition type (ferromagnetic Curie, antiferromagnetic Néel, ferroelectric, metal-insulator, charge-density-wave, structural) plus an ungated melting-point regressor as a universal calibration anchor, at an $\sim 8\%$ parameter increase ($\sim 9\text{M}$ added parameters) with transfer learning from the final model. The Curie MVP is trained; the expanded corpus already stages the next labels (10,039 Néel, 4,701 Debye, 2,113 melting-point examples); a quantile-regression upgrade for calibrated uncertainty intervals is designed. These are roadmap items, not results.

The data engine. The binding constraint on that roadmap is labels: Néel temperatures, Debye temperatures, and pressure-dependent transitions barely exist in curated databases—they live in the text of papers. To grow the dataset at scale, I built an agentic literature-curation pipeline—the *data engine* for Hydra MAGE—on Claude models via the agent SDK: six specialized agents (extraction, unit normalization, formula canonicalization, deduplication, physics-plausibility filtering, and a sole-writer ingester) coordinate through 13 typed tools against a knowledge graph whose every write emits a SHA-256-signed event into an append-only audit table. A verification layer (read-after-write, idempotency, and deliberate-corruption “sabotage” tests, 19/19 passing) guards data integrity. On a real-corpus smoke test over 55 recent arXiv cond-mat .supr-con abstracts, the pipeline ingested 3 materials with full provenance (including a pressure-dependent Hf-doped ZrTe_3 and Hg-1223 at ambient and elevated pressure), correctly identified 40 abstracts as carrying no extractable property claim, and escalated 12 ambiguous cases to human review; a fast mode achieves $14.8\times$ the throughput at $4.8\times$ lower cost per record. Honest status: the pipeline is built and validated end-to-end, but has *not yet been run at web scale*—the deferral is purely API cost, not capability.

Other modeling directions. Reducing residual latent torsion through explicit manifold alignment losses (penalizing T_c error at Z-vectors that produce high-similarity formulas) or self-consistency training—noting that extended training alone already contracted torsion $\sim 3\times$; incorporating physical constraints (charge neutrality, stability) in generation—now partly realized by the validity-head successor of Section 4.8, whose physical-possibility head scores chemical admissibility directly; extending the external validation of Section 4.8 to further curated databases (SODNet); and addressing the stoichiometric precision bottleneck through constrained decoding or post-hoc normalization.

6 Conclusion

I developed a multi-task sequence encoder-decoder that achieves $R^2 = 0.9999$ on T_c prediction—a $\sim 30\times$ error reduction over a composition-only baseline—and recovers 429 never-seen superconductors from disjoint databases, scored jointly on composition *and* T_c , through a Z-space search over generated formulas—the base model reaching 423/429 and a validity-head successor of near-identical latent geometry (Section 4.8) closing the remainder to 429/429 (the last target via T_c -gradient steering)—with zero hallucinated elements and a mean absolute T_c error of 0.26 K (median 0.0 K; tail to ~ 13 K on the hardest multi-dopant cuprates), all without receiving any target information.

Then I opened the model up. Symbolic regression distills its T_c head into per-family closed-form equations (per-family R^2 0.65–0.98; quadratic or exponential forms across every cuprate family)—surviving a contamination retraction that is itself a result about how such extractions fail. Gradient censuses locate the computation: T_c lives in 2–10 principal directions per family at 90% variance,

reconstruction competence in exactly one, element identities correlated with nameable dimensions (oxygen at $r = +0.83$ on dimension 276, copper bipolar at ± 0.70)—though causal intervention (Section 4.10) reveals these are *distributed* directions, not individually causal dimensions, a superposition result outside the language-model setting where it is usually studied. The model exhibits an emergent phase-transition energy scale it was never trained on—predicting Curie temperatures of never-seen ferromagnets with perfect rank order—which a trained Curie head converts into near-SOTA performance and a roadmap toward a universal phase-transition model, with an agentic literature-curation pipeline built as its data engine. And the interpretability artifacts steer generation: gradient ascent constrained by per-head subspaces, dead-dimension masks, and element-correlation vectors, filtered by a deterministic chemistry gate, economic viability, and a $\sim 365\text{K}$ -entry novelty check. This pipeline yields a cross-family shortlist whose per-candidate literature search returns mostly rediscoveries of known superconducting chemistry and a single deeply-vetted novel prediction—Hf-doped FeSe ($\sim 21\text{ K}$ predicted), the one carried through a full literature search—itsself a hypothesis awaiting experimental vetting.

The model has learned what I characterize as a 2048-parameter effective theory of superconductivity—a single representation spanning 14 material families that encodes family boundaries, the SC/non-SC phase boundary, and continuous T_c gradients within a geometrically organized manifold. The contribution of this paper is that the theory is no longer only a characterization: parts of it have been extracted as equations, its anatomy mapped, its emergent reach documented, and its structure used for control.

Code is available at (73).

References

- [1] J. G. Bednorz and K. A. Müller, “Possible high T_c superconductivity in the Ba-La-Cu-O system,” *Z. Phys. B*, vol. 64, pp. 189–193, 1986.
- [2] Y. Kamihara, T. Watanabe, M. Hirano, and H. Hosono, “Iron-based layered superconductor $\text{La}[\text{O}_{1-x}\text{F}_x]\text{FeAs}$ ($x = 0.05\text{--}0.12$) with $T_c = 26\text{ K}$,” *J. Am. Chem. Soc.*, vol. 130, pp. 3296–3297, 2008.
- [3] J. Nagamatsu, N. Nakagawa, T. Muranaka, Y. Zenitani, and J. Akimitsu, “Superconductivity at 39 K in magnesium diboride,” *Nature*, vol. 410, pp. 63–64, 2001.
- [4] A. P. Drozdov, M. I. Erements, I. A. Troyan, V. Ksenofontov, and S. I. Shylin, “Conventional superconductivity at 203 kelvin at high pressures in the sulfur hydride system,” *Nature*, vol. 525, pp. 73–76, 2015.
- [5] M. Somayazulu *et al.*, “Evidence for superconductivity above 260 K in lanthanum superhydride at megabar pressures,” *Phys. Rev. Lett.*, vol. 122, p. 027001, 2019.
- [6] W. E. Pickett, “Colloquium: Room temperature superconductivity: The roles of theory and materials design,” *Rev. Mod. Phys.*, vol. 95, p. 021001, 2023.
- [7] J. Bardeen, L. N. Cooper, and J. R. Schrieffer, “Theory of superconductivity,” *Phys. Rev.*, vol. 108, no. 5, pp. 1175–1204, 1957.
- [8] M. Tinkham, *Introduction to Superconductivity*, 2nd ed. New York: Dover, 2004.
- [9] Z. Fan, J.-X. Zhang, B. Zhan, D. Lv, and X.-Y. Song, “Intrinsic constraint on T_c for unconventional superconductivity,” *npj Quantum Materials*, vol. 10, p. 11, 2025.
- [10] P. W. Anderson, “Plasmons, gauge invariance, and mass,” *Phys. Rev.*, vol. 130, no. 1, pp. 439–442, 1963.
- [11] K. Hamidieh, “A data-driven statistical model for predicting the critical temperature of a superconductor,” *Computational Materials Science*, vol. 154, pp. 346–354, 2018.
- [12] V. Stanev, C. Oses, A. G. Kusne, E. Rodriguez, J. Paglione, S. Curtarolo, and I. Takeuchi, “Machine learning modeling of superconducting critical temperature,” *npj Computational Materials*, vol. 4, p. 28, 2018.

- [13] L. Ward, A. Agrawal, A. Choudhary, and C. Wolverton, “A general-purpose machine learning framework for predicting properties of inorganic materials,” *npj Computational Materials*, vol. 2, p. 16028, 2016.
- [14] S. G. Jung, G. Jung, and J. M. Cole, “Machine-learning predictions of critical temperatures from chemical compositions of superconductors,” *J. Chem. Inf. Model.*, 2024.
- [15] T. Sommer, R. Willa, J. Schmalian, and P. Friederich, “3DSC—a dataset of superconductors including crystal structures,” *Scientific Data*, vol. 10, p. 816, 2023.
- [16] J. A. Slezak *et al.*, “Imaging the impact on cuprate superconductivity of varying the interatomic distances within individual crystal unit cells,” *PNAS*, vol. 105, pp. 3203–3208, 2008.
- [17] E. Pavarini, I. Dasgupta, T. Saha-Dasgupta, O. Jepsen, and O. K. Andersen, “Band-structure trend in hole-doped cuprates and correlation with $T_{c,max}$,” *Phys. Rev. Lett.*, vol. 87, p. 047003, 2001.
- [18] D. Wines, T. Xie, and K. Choudhary, “Inverse design of next-generation superconductors using data-driven deep generative models,” *J. Phys. Chem. Lett.*, vol. 14, pp. 6630–6638, 2023.
- [19] C. Zhong *et al.*, “Deep generative model for inverse design of high-temperature superconductor compositions with predicted $T_c > 77$ K,” *ACS Appl. Mater. Interfaces*, vol. 15, pp. 30029–30038, 2023.
- [20] “ScGAN: A generative adversarial network to predict hypothetical superconductors,” *J. Phys.: Condens. Matter*, vol. 36, p. 025901, 2023.
- [21] D. P. Kingma and M. Welling, “Auto-encoding variational Bayes,” in *Proc. ICLR*, 2014.
- [22] Z. Liu, M. Jiang, and T. Luo, “Machine learning prediction of superconducting critical temperature through the structural descriptor,” *J. Phys. Chem. C*, vol. 126, pp. 8922–8931, 2022.
- [23] S. Yuan and S. V. Dordevic, “Diffusion models for conditional generation of hypothetical new families of superconductors,” *Scientific Reports*, vol. 14, p. 10275, 2024.
- [24] E. Kim, J. Noh, G. H. Gu, A. Aspuru-Guzik, and Y. Jung, “Generative adversarial networks for crystal structure prediction,” *ACS Central Science*, vol. 6, no. 8, pp. 1412–1420, 2020.
- [25] R. Gómez-Bombarelli *et al.*, “Automatic chemical design using a data-driven continuous representation of molecules,” *ACS Central Science*, vol. 4, no. 2, pp. 268–276, 2018.
- [26] J. Noh, J. Kim, H. S. Stein, B. Sanchez-Lengeling, J. M. Gregoire, A. Aspuru-Guzik, and Y. Jung, “Inverse design of solid-state materials via a continuous representation,” *Matter*, vol. 1, no. 5, pp. 1370–1384, 2019.
- [27] Y. Dan, Y. Zhao, X. Li, S. Li, M. Hu, and J. Hu, “Generative adversarial networks (GAN) based efficient sampling of chemical composition space for inverse design of inorganic materials,” *npj Computational Materials*, vol. 6, no. 1, p. 84, 2020.
- [28] A. See, P. J. Liu, and C. D. Manning, “Get to the point: Summarization with pointer-generator networks,” in *Proc. ACL*, pp. 1073–1083, 2017.
- [29] T. Bricken, A. Templeton, J. Batson, et al., “Towards monosemanticity: Decomposing language models with dictionary learning,” *Transformer Circuits Thread*, 2023.
- [30] A. Templeton, T. Conerly, J. Marcus, et al., “Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet,” *Transformer Circuits Thread*, 2024.
- [31] S. Marks, C. Rager, E. J. Michaud, Y. Belinkov, D. Bau, and A. Mueller, “Sparse feature circuits: discovering and editing interpretable causal graphs in language models,” *Proc. Int. Conf. Learning Representations (ICLR)*, 2025.

- [32] A. Zou, L. Phan, S. Chen, et al., “Representation engineering: a top-down approach to AI transparency,” *arXiv:2310.01405*, 2023.
- [33] A. M. Turner, L. Thiergart, G. Leech, D. Udell, J. J. Vazquez, U. Mini, and M. MacDiarmid, “Activation addition: steering language models without optimization,” *arXiv:2308.10248*, 2023.
- [34] K. Li, A. K. Hopkins, D. Bau, F. Viégas, H. Pfister, and M. Wattenberg, “Emergent world representations: exploring a sequence model trained on a synthetic task,” *Proc. Int. Conf. Learning Representations (ICLR)*, 2023.
- [35] J. Hewitt and P. Liang, “Designing and interpreting probes with control tasks,” *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, pp. 2733–2743, 2019.
- [36] P. Quirke, “Ablation-reversible heads don’t transfer: a stress test for mechanistic role claims in transformers,” *arXiv:2606.08292*, 2026.
- [37] G. Minegishi, H. Furuta, T. Kojima, Y. Iwasawa, and Y. Matsuo, “Understanding emergent misalignment via feature superposition geometry,” *arXiv:2605.00842*, 2026.
- [38] D. Pluth, Z. N. Houghton, Y. Zhou, and V. K. Gurbani, “Mechanistic interpretability of ASR models using sparse autoencoders,” *arXiv:2605.12225*, 2026.
- [39] M. Cranmer, “Interpretable machine learning for science with PySR and SymbolicRegression.jl,” *arXiv:2305.01582*, 2023.
- [40] K. Vafa, P. G. Chang, A. Rambachan, and S. Mullainathan, “What has a foundation model found? Using inductive bias to probe for world models,” in *Proc. ICML*, 2025.
- [41] A. Merchant, S. Batzner, S. S. Schoenholz, M. Aykol, G. Cheon, and E. D. Cubuk, “Scaling deep learning for materials discovery,” *Nature*, vol. 624, pp. 80–85, 2023.
- [42] D. W. Davies, K. T. Butler, A. J. Jackson, A. Morris, J. M. Frost, J. M. Skelton, and A. Walsh, “Computational Screening of All Stoichiometric Inorganic Materials,” *Chem*, vol. 1, no. 4, pp. 617–627, 2016. DOI: 10.1016/j.chempr.2016.09.010.
- [43] S. G. Jung, G. Jung, and J. M. Cole, “Machine-learning prediction of Curie temperature from chemical compositions of ferromagnetic materials,” *J. Chem. Inf. Model.*, vol. 64, no. 16, pp. 6388–6409, 2024. DOI: 10.1021/acs.jcim.4c00947.
- [44] SuperCon Database, National Institute for Materials Science (NIMS), Japan. DOI: 10.48505/nims.4487.
- [45] H. Saini *et al.*, “NEMAD: A database of novel and emerging materials,” *npj Computational Materials*, 2024.
- [46] K. Choudhary *et al.*, “The joint automated repository for various integrated simulations (JARVIS) for data-driven materials design,” *npj Computational Materials*, vol. 6, p. 173, 2020.
- [47] A. Jain *et al.*, “The Materials Project: A materials genome approach to accelerating materials innovation,” *APL Materials*, vol. 1, no. 1, p. 011002, 2013.
- [48] S. P. Ong *et al.*, “Python Materials Genomics (pymatgen): A robust, open-source python library for materials analysis,” *Computational Materials Science*, vol. 68, pp. 314–319, 2013.
- [49] L. Ward *et al.*, “Matminer: An open source toolkit for materials data mining,” *Computational Materials Science*, vol. 152, pp. 60–69, 2018.
- [50] R. T. Mullapudi, W. R. Mark, N. Shazeer, and K. Fatahalian, “HydraNets: Specialized dynamic architectures for efficient inference,” in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, pp. 8080–8089, 2018.

- [51] H. W. Kuhn, “The Hungarian method for the assignment problem,” *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955.
- [52] A. Vaswani *et al.*, “Attention is all you need,” in *Proc. NeurIPS*, pp. 5998–6008, 2017.
- [53] P. B. Allen and R. C. Dynes, “Transition temperature of strong-coupled superconductors reanalyzed,” *Phys. Rev. B*, vol. 12, no. 3, pp. 905–922, 1975.
- [54] M. R. Presland, J. L. Tallon, R. G. Buckley, R. S. Liu, and N. E. Flower, “General trends in oxygen stoichiometry effects on T_c in Bi and Tl superconductors,” *Physica C*, vol. 176, pp. 95–105, 1991.
- [55] A. Paszke *et al.*, “PyTorch: An imperative style, high-performance deep learning library,” in *NeurIPS*, 2019.
- [56] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proc. ICCV*, pp. 2980–2988, 2017.
- [57] S. J. Rennie, E. Marcheret, Y. Mroueh, J. Ross, and V. Goel, “Self-critical sequence training for image captioning,” in *Proc. CVPR*, pp. 7008–7024, 2017.
- [58] A. A. Abrikosov, “On the magnetic properties of superconductors of the second group,” *Sov. Phys. JETP*, vol. 5, pp. 1174–1182, 1957.
- [59] L. McInnes, J. Healy, and J. Melville, “UMAP: Uniform manifold approximation and projection for dimension reduction,” *arXiv:1802.03426*, 2018.
- [60] L. van der Maaten and G. Hinton, “Visualizing data using t-SNE,” *Journal of Machine Learning Research*, vol. 9, pp. 2579–2605, 2008.
- [61] L. McInnes, J. Healy, and S. Astels, “hdbscan: Hierarchical density based clustering,” *Journal of Open Source Software*, vol. 2, no. 11, p. 205, 2017.
- [62] X. Luo *et al.*, “Electronic origin of high superconducting critical temperature in trilayer cuprates,” *Nature Physics*, vol. 19, pp. 1841–1847, 2023.
- [63] M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, 2nd ed. New York: Dover, 2016.
- [64] F. R. Fickett, “Standards for measurement of the critical fields of superconductors,” *J. Res. Natl. Bur. Stand.*, vol. 90, no. 2, pp. 95–128, 1985.
- [65] L. Foppiano *et al.*, “SuperMat: construction of a linked annotated dataset from superconductors-related publications,” *Science and Technology of Advanced Materials: Methods*, vol. 1, no. 1, pp. 34–44, 2021.
- [66] A. Geiger, H. Lu, T. Icard, and C. Potts, “Causal abstractions of neural networks,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [67] K. Wang, A. Variengien, A. Conmy, B. Shlegeris, and J. Steinhardt, “Interpretability in the wild: a circuit for indirect object identification in GPT-2 small,” *Proc. Int. Conf. Learning Representations (ICLR)*, 2023.
- [68] L. Chan, A. Garriga-Alonso, N. Goldowsky-Dill, R. Greenblatt, J. Nitishinskaya, A. Radhakrishnan, B. Shlegeris, and N. Thomas, “Causal scrubbing: a method for rigorously testing interpretability hypotheses,” *AI Alignment Forum*, 2022.
- [69] A. Nguyen, J. Yosinski, and J. Clune, “Deep neural networks are easily fooled: high confidence predictions for unrecognizable images,” *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, pp. 427–436, 2015.
- [70] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” *Proc. Int. Conf. Machine Learning (ICML)*, pp. 1321–1330, 2017.

- [71] G. Alain and Y. Bengio, “Understanding intermediate layers using linear classifier probes,” *arXiv:1610.01644*, 2016.
- [72] Y. Belinkov, “Probing classifiers: promises, shortcomings, and advances,” *Computational Linguistics*, vol. 48, no. 1, pp. 207–219, 2022.
- [73] J. Conde, “Hydra MAGE: a many-headed Materials Autoencoder for Generative Exploration,” 2026. [Online]. Available: <https://github.com/jamesconde/superconductor-vae>
- [74] F. C. Hsu, J. Y. Luo, K. W. Yeh, T. K. Chen, T. W. Huang, P. M. Wu, Y. C. Lee, Y. L. Huang, Y. Y. Chu, D. C. Yan, and M. K. Wu, “Superconductivity in the PbO-type structure α -FeSe,” *Proc. Natl. Acad. Sci. U.S.A.*, vol. 105, no. 38, pp. 14262–14264, 2008. DOI: 10.1073/pnas.0807325105
- [75] Q. Y. Wang, Z. Li, W. H. Zhang, *et al.*, “Interface-induced high-temperature superconductivity in single unit-cell FeSe films on SrTiO₃,” *Chin. Phys. Lett.*, vol. 29, no. 3, art. 037402, 2012. DOI: 10.1088/0256-307X/29/3/037402
- [76] S. He, J. He, W. Zhang, *et al.*, “Phase diagram and electronic indication of high-temperature superconductivity at 65 K in single-layer FeSe films,” *Nat. Mater.*, vol. 12, pp. 605–610, 2013. DOI: 10.1038/nmat3648
- [77] A. K. Yadav, A. V. Sanchela, A. D. Thakur, and C. V. Tomy, “Effect of nominal substitution of transition metals for excess Fe in Fe_{1+x}Se superconductor,” *Solid State Commun.*, vol. 202, pp. 8–13, 2015. DOI: 10.1016/j.ssc.2014.10.027
- [78] Y. Mizuguchi, F. Tomioka, S. Tsuda, T. Yamaguchi, and Y. Takano, “Substitution effects on FeSe superconductor,” *J. Phys. Soc. Jpn.*, vol. 78, no. 7, art. 074712, 2009. DOI: 10.1143/JPSJ.78.074712
- [79] A. F. Wells, *Structural Inorganic Chemistry*, 5th ed. Oxford: Clarendon Press, 1984.
- [80] H. Kamerlingh Onnes, “Further experiments with liquid helium. IV. The resistance of pure mercury at helium temperatures,” *Commun. Phys. Lab. Univ. Leiden*, no. 120b, 1911.
- [81] X. Lai, H. Zhang, Y. Wang, X. Wang, X. Zhang, J. Lin, and F. Huang, “Observation of superconductivity in tetragonal FeS,” *J. Am. Chem. Soc.*, vol. 137, no. 32, pp. 10148–10151, 2015. DOI: 10.1021/jacs.5b06687.
- [82] J.-F. Ge, Z.-L. Liu, C. Liu, C.-L. Gao, D. Qian, Q.-K. Xue, Y. Liu, and J.-F. Jia, “Superconductivity above 100 K in single-layer FeSe films on doped SrTiO₃,” *Nat. Mater.*, vol. 14, no. 3, pp. 285–289, 2015. DOI: 10.1038/nmat4153.
- [83] M. Rotter, M. Tegel, and D. Johrendt, “Superconductivity at 38 K in the iron arsenide (Ba_{1-x}K_x)Fe₂As₂,” *Phys. Rev. Lett.*, vol. 101, no. 10, art. 107006, 2008. DOI: 10.1103/PhysRevLett.101.107006.
- [84] A. P. Ramirez, R. J. Cava, and J. Krajewski, “Colossal magnetoresistance in Cr-based chalcogenide spinels,” *Nature*, vol. 386, pp. 156–159, 1997. DOI: 10.1038/386156a0.
- [85] S. Doniach and S. Engelsberg, “Low-temperature properties of nearly ferromagnetic Fermi liquids,” *Phys. Rev. Lett.*, vol. 17, no. 14, pp. 750–753, 1966. DOI: 10.1103/PhysRevLett.17.750.
- [86] U.S. Geological Survey, *Mineral Commodity Summaries 2025*. Reston, VA: U.S. Geological Survey, 2025. DOI: 10.3133/mcs2025.
- [87] A. S. Sefat, R. Jin, M. A. McGuire, B. C. Sales, D. J. Singh, and D. Mandrus, “Superconductivity at 22 K in Co-doped BaFe₂As₂ crystals,” *Phys. Rev. Lett.*, vol. 101, no. 11, art. 117004, 2008. DOI: 10.1103/PhysRevLett.101.117004.
- [88] B. T. Matthias, R. M. Bozorth, and J. H. Van Vleck, “Ferromagnetic interaction in EuO,” *Phys. Rev. Lett.*, vol. 7, no. 5, pp. 160–161, 1961. DOI: 10.1103/PhysRevLett.7.160.

- [89] H. Ohno, “Making nonmagnetic semiconductors ferromagnetic,” *Science*, vol. 281, no. 5379, pp. 951–956, 1998. DOI: 10.1126/science.281.5379.951.
- [90] E. Liu, Y. Sun, N. Kumar, *et al.*, “Giant anomalous Hall effect in a ferromagnetic kagome-lattice semimetal,” *Nat. Phys.*, vol. 14, no. 11, pp. 1125–1131, 2018. DOI: 10.1038/s41567-018-0234-5.